Basic Definitions

Alexander Golovnev

Outline

The Degree of a Vertex

Paths

Connectivity

Directed Graphs

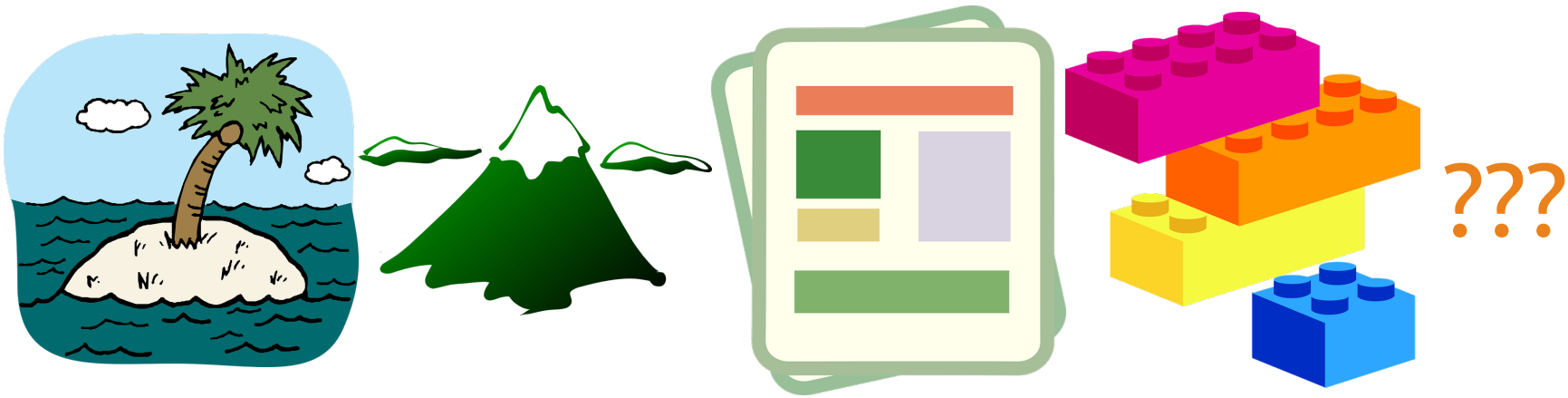
Weighted Graphs

Definitions

An isolated vertex forms a component

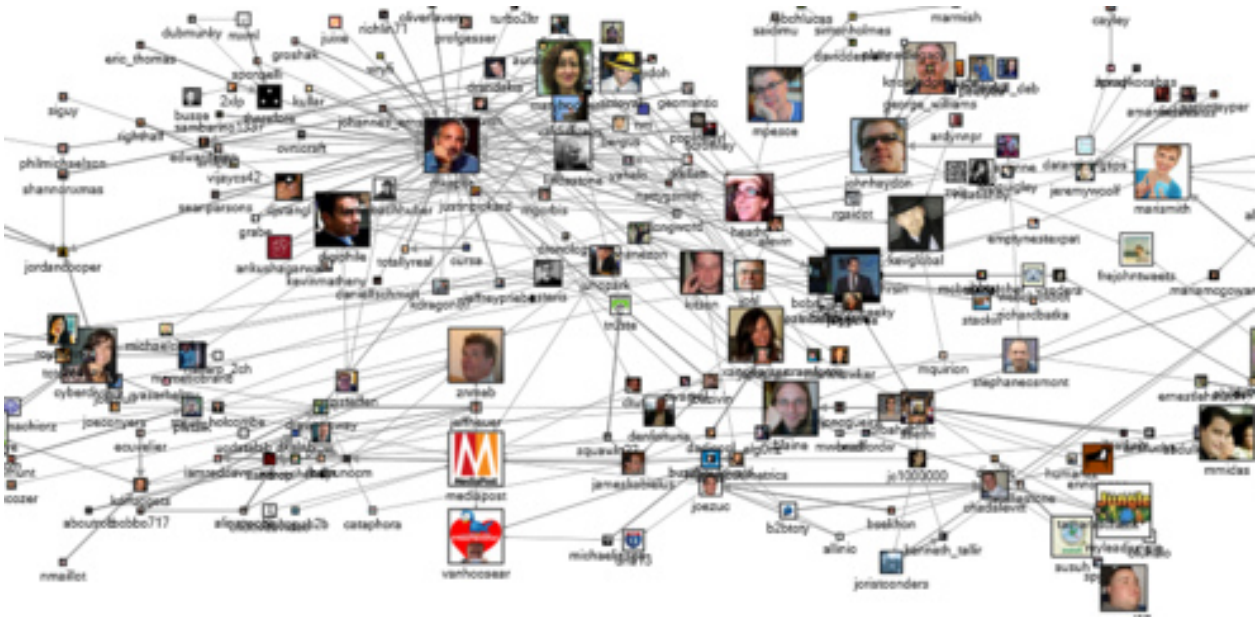
Definitions

An isolated vertex forms a component



The Degree of a Vertex

The number of friends



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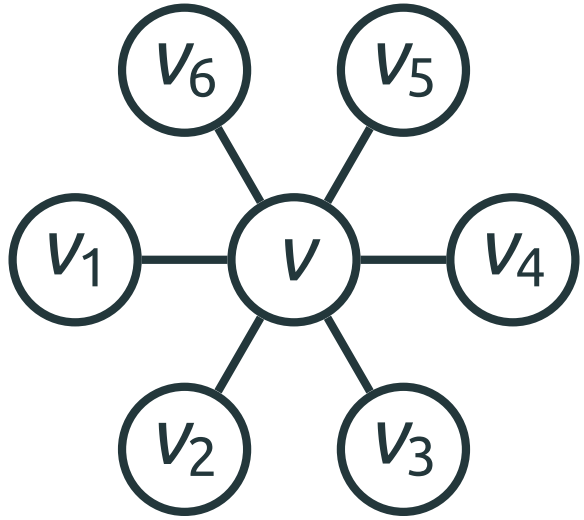
The Degree of a Vertex

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- I.e., the Degree of a vertex is the number of its neighbors
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- The degree of a graph is the maximum degree of its vertices

The Degree of a Vertex: Examples

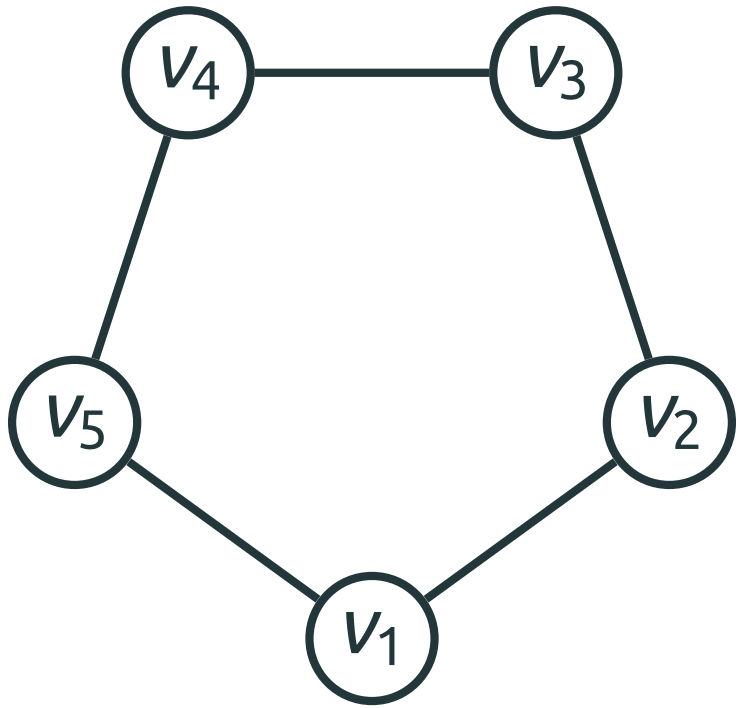
The degree of v is 6: $\deg(v) = 6$

The degree of v_6 is 1: $\deg(v_6) = 1$

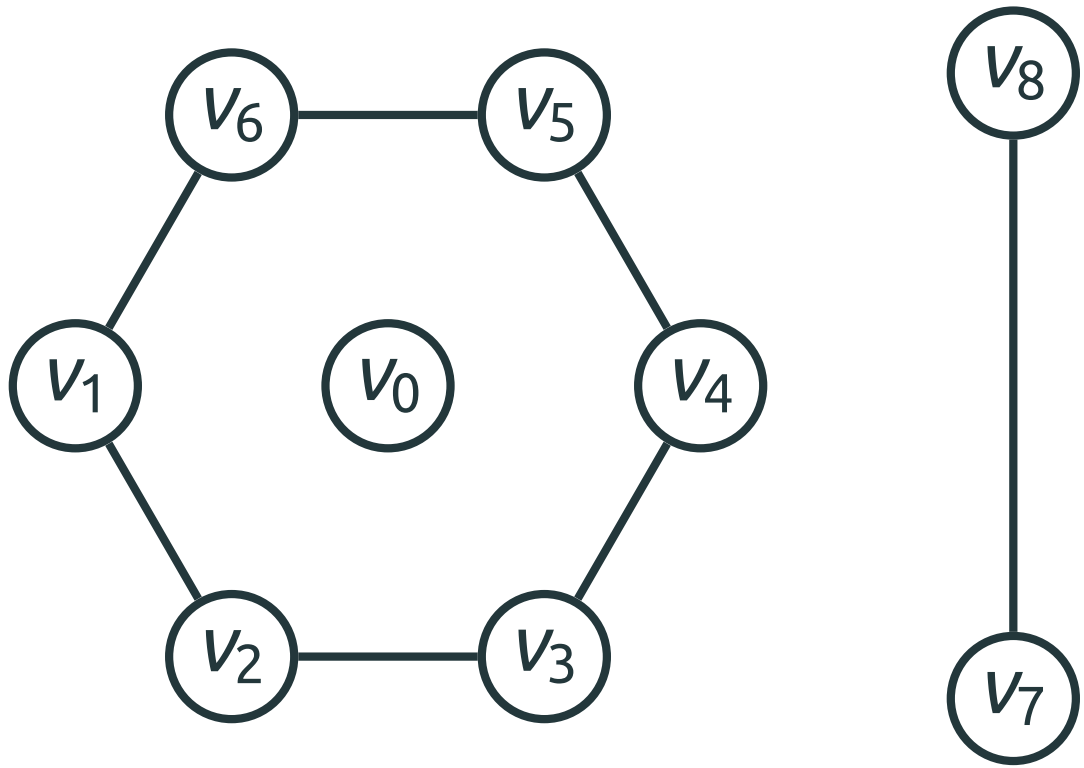


The Degree of a Vertex: Examples

The degree of every vertex is 2: $\forall i, \deg(v_i) = 2$

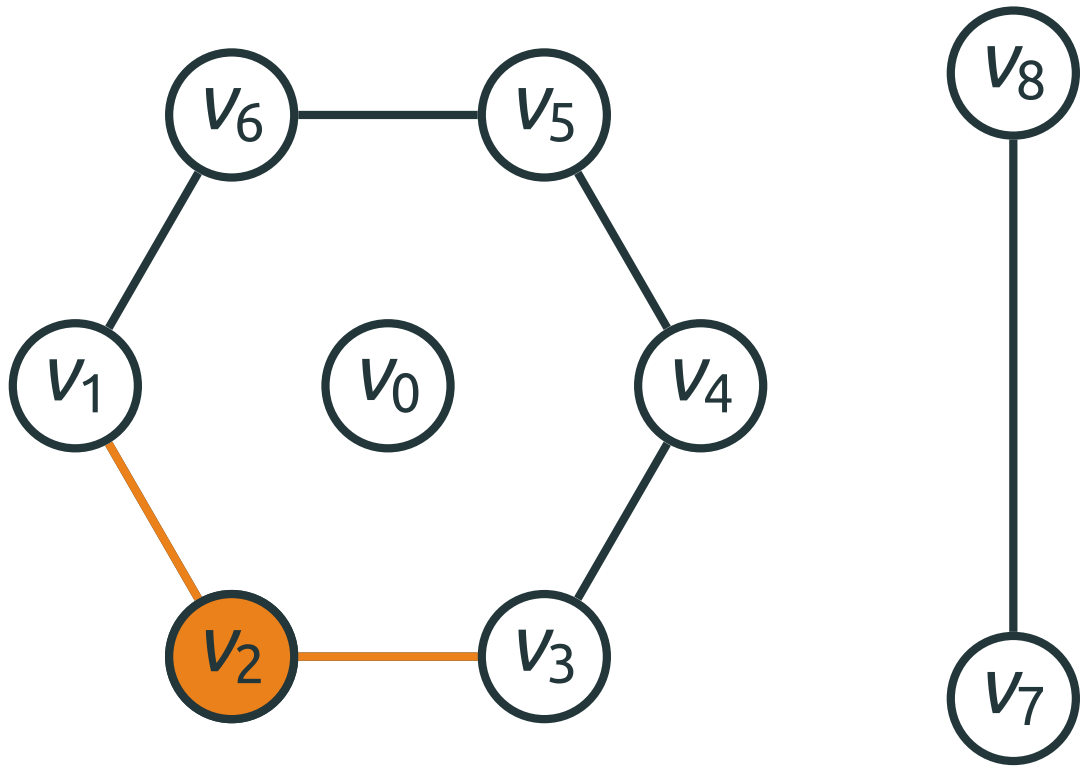


Isolated Vertices



Isolated Vertices

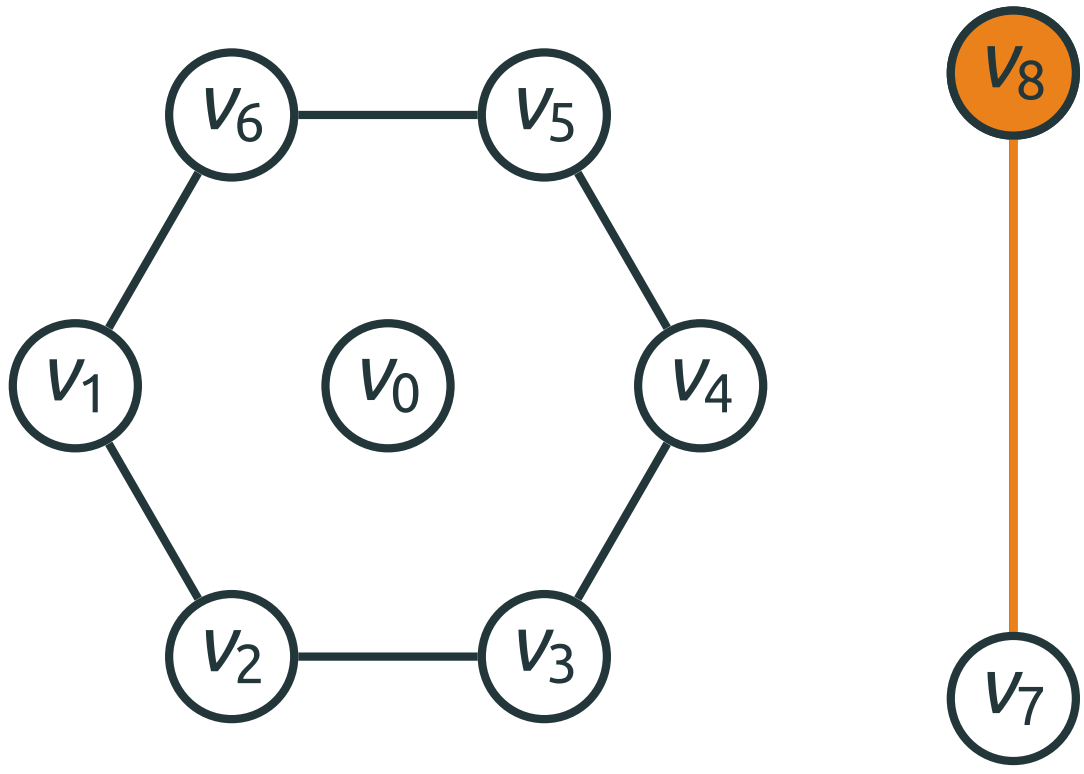
$\deg(v_2) = 2$



Isolated Vertices

$\deg(v_2) = 2$

$\deg(v_8) = 1$

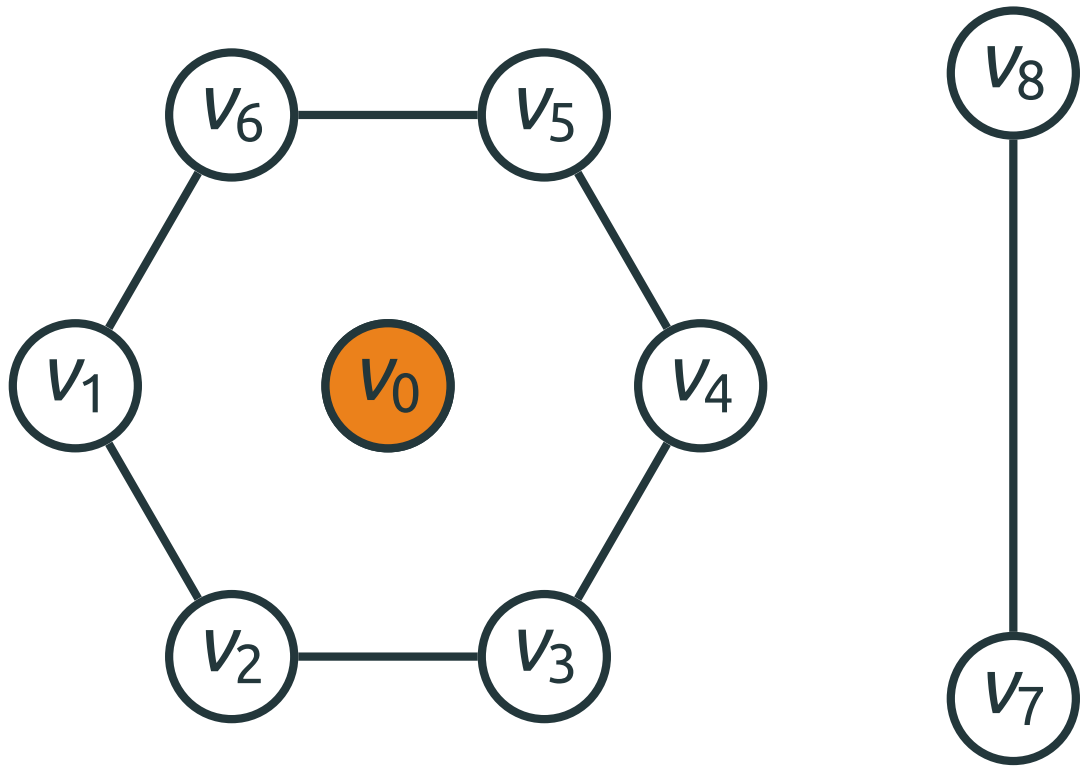


Isolated Vertices

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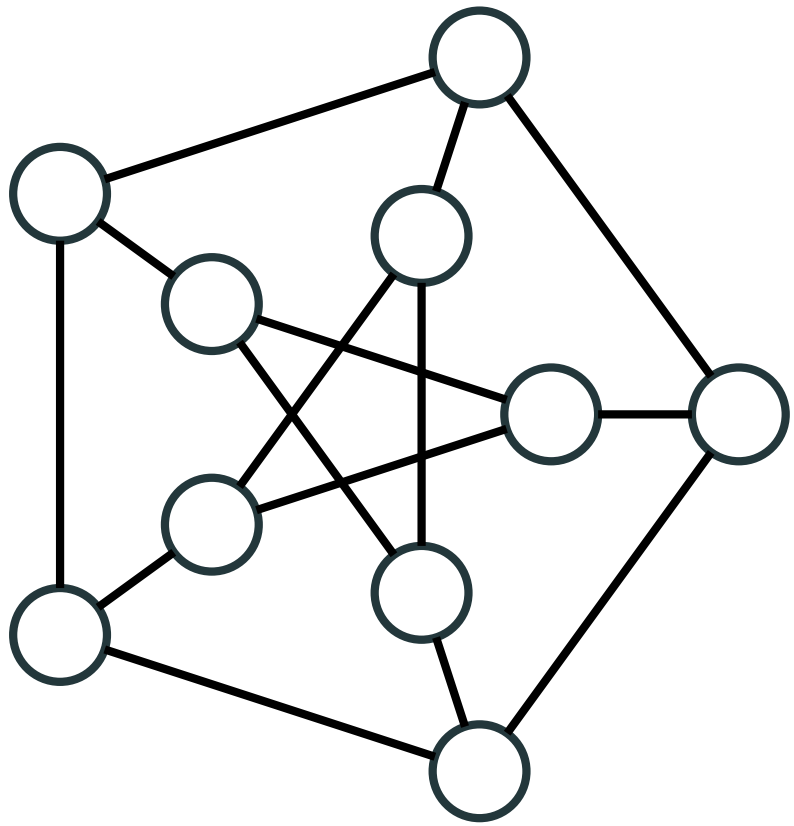
$\deg(v_8) = 1$

$\deg(v_0) = 0$. v_0 is an Isolated Vertex



Regular Graphs

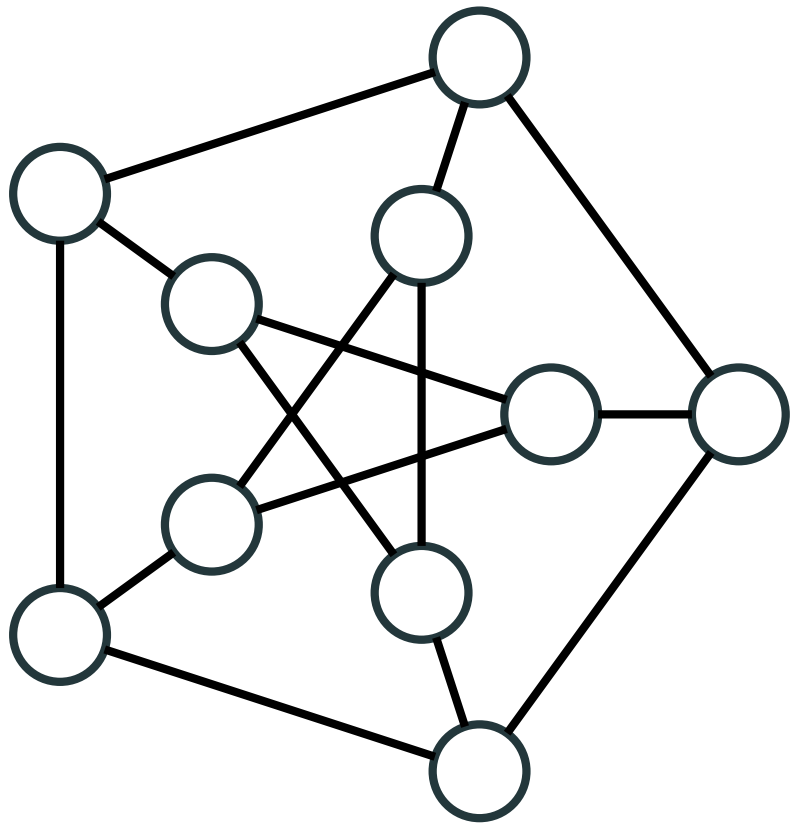
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A regular graph of degree *k* is also called *k*-Regular

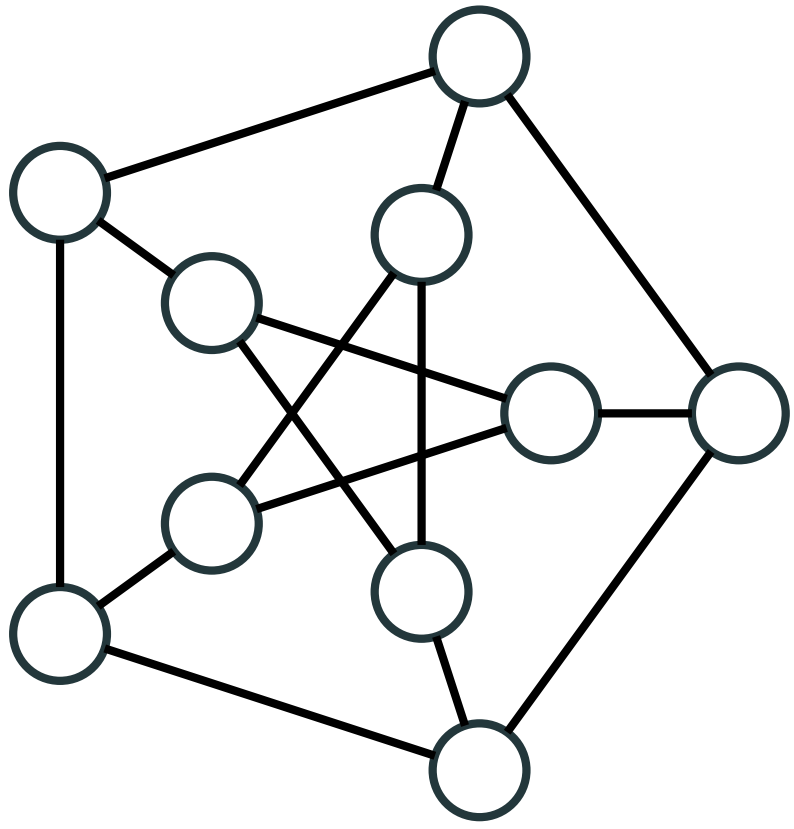


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E.g., this graph is 3-Regular



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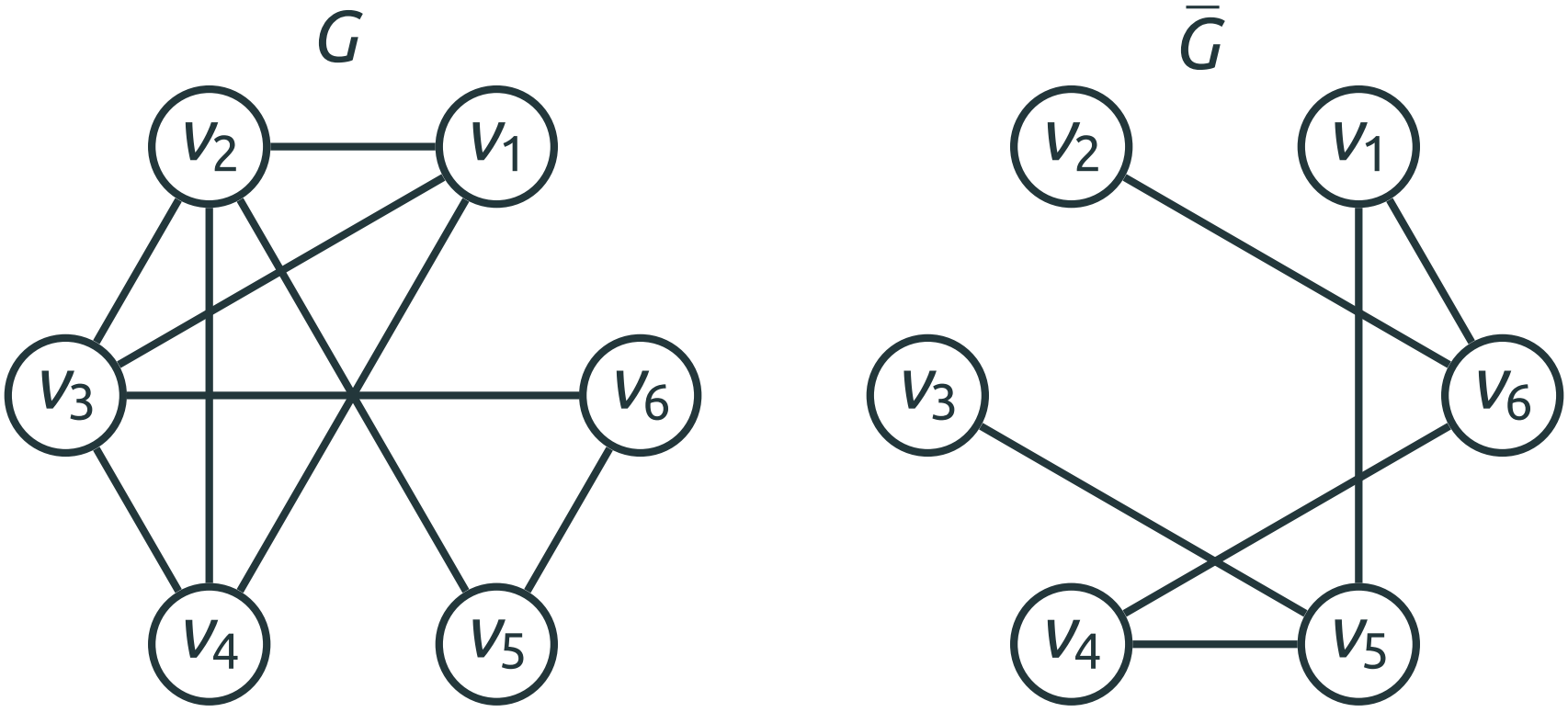
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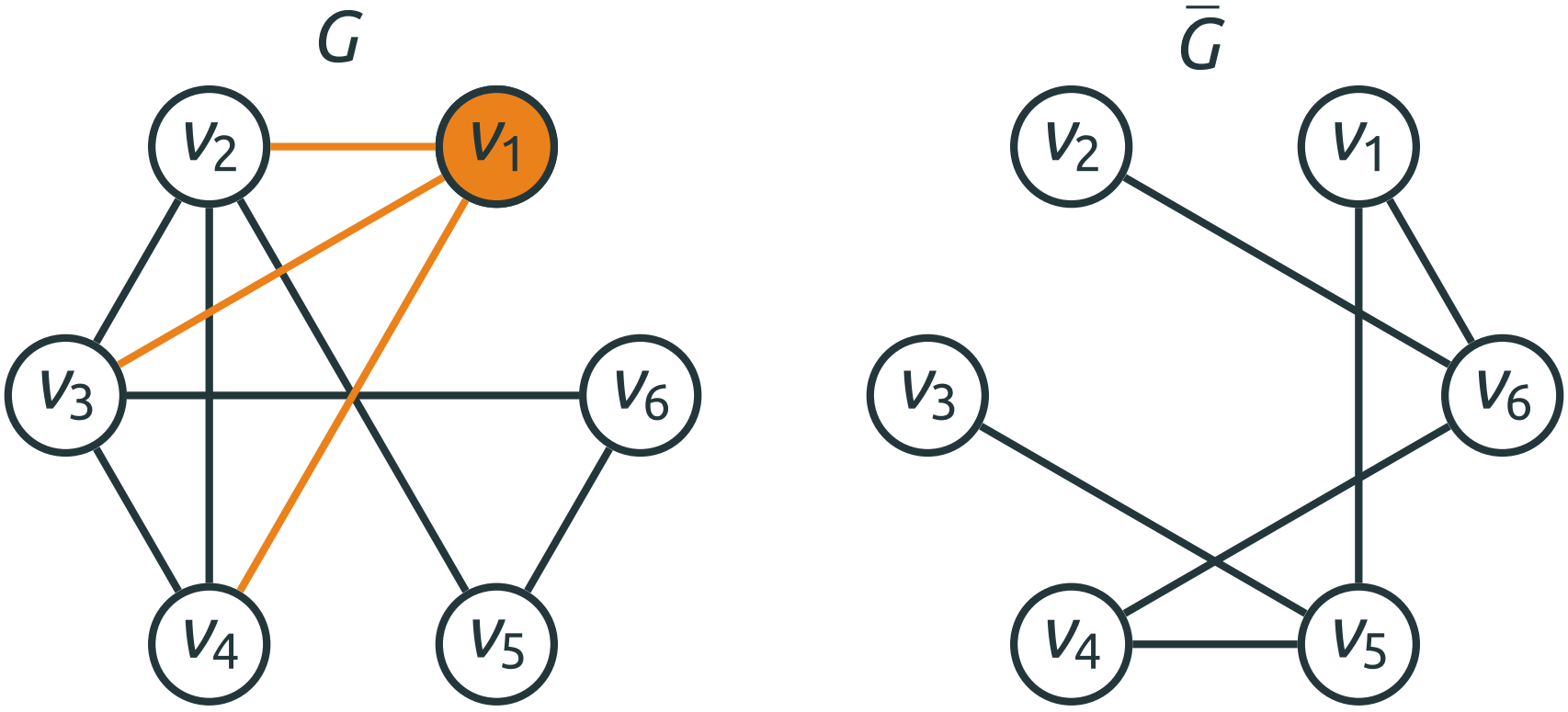
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- I.e., $(u, v) \in \bar{E}$ if and only if $(u, v) \notin E$

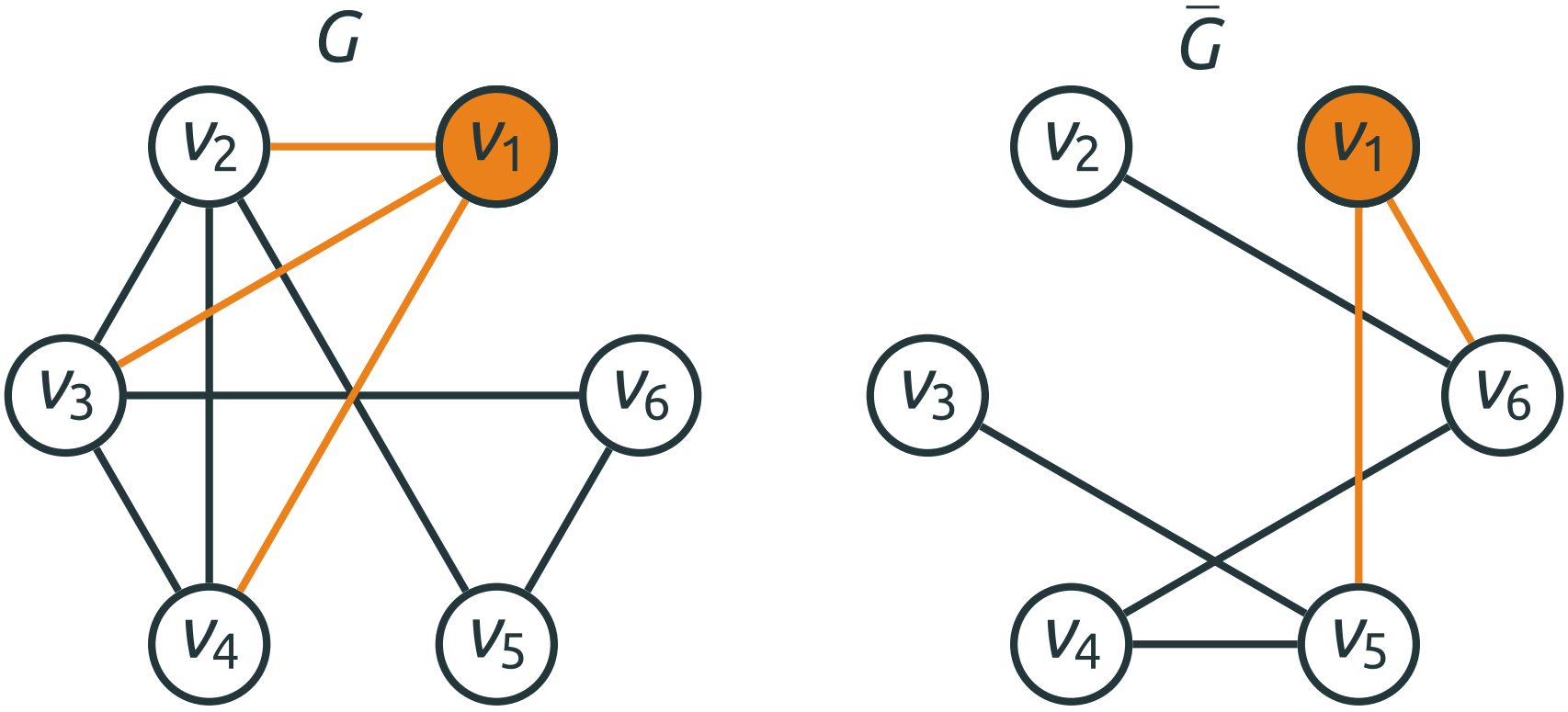
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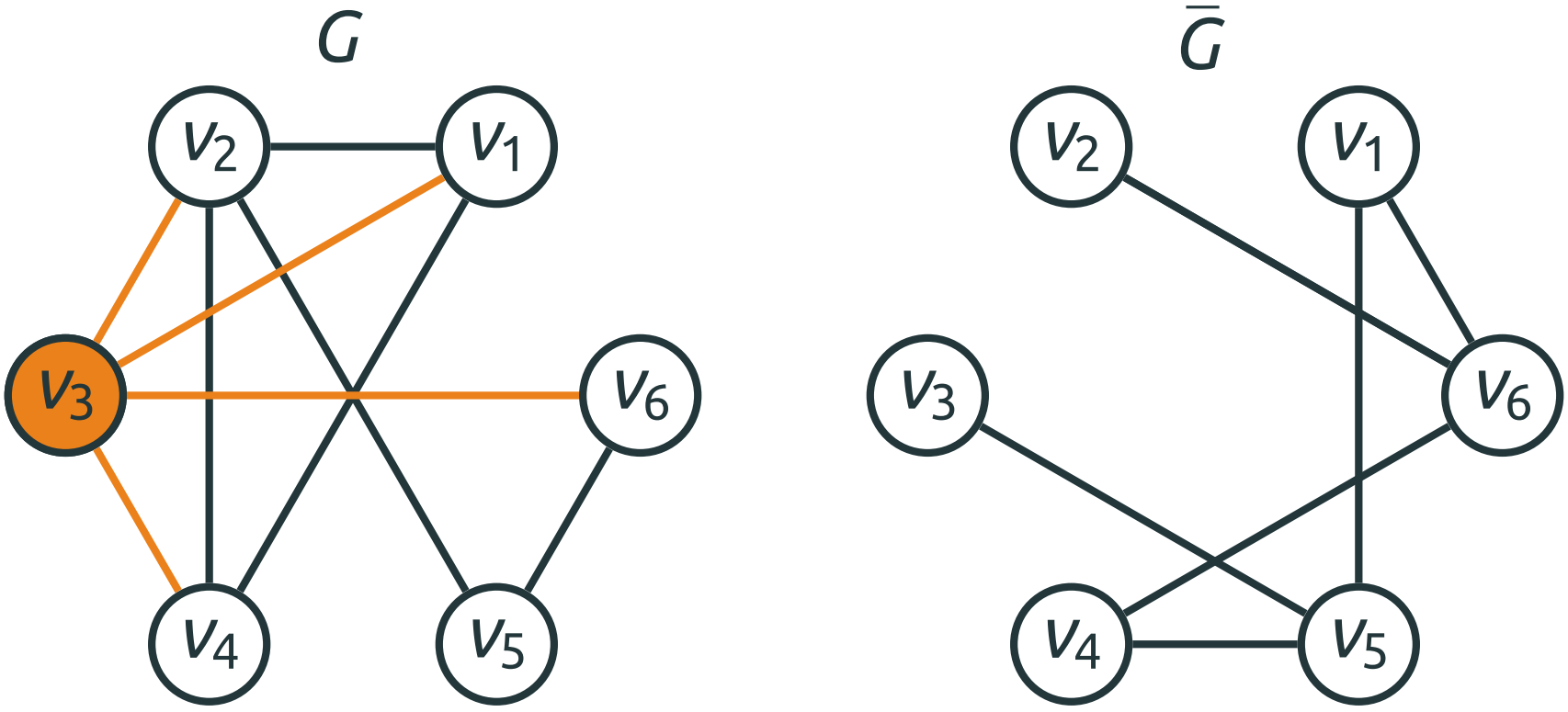
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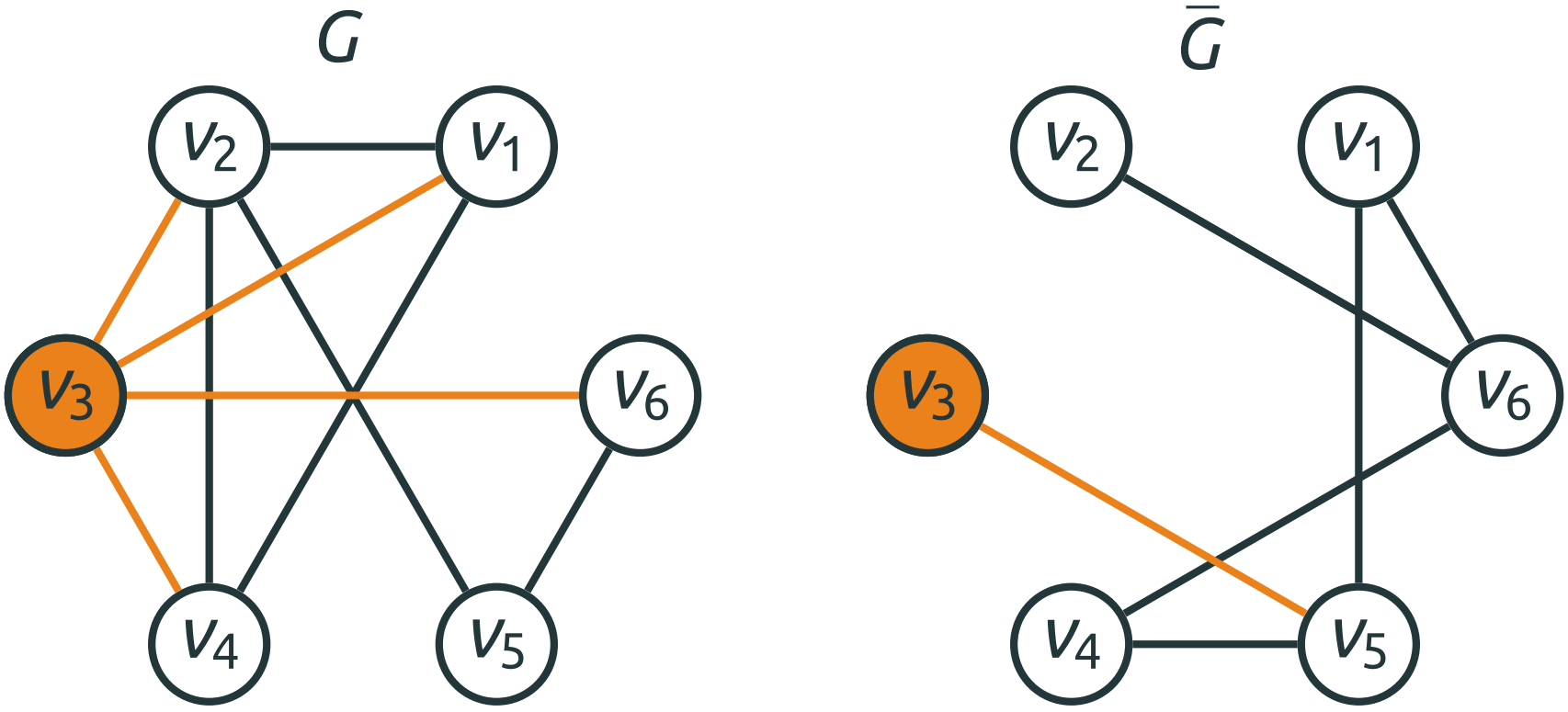
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Is there a path from one point to another?



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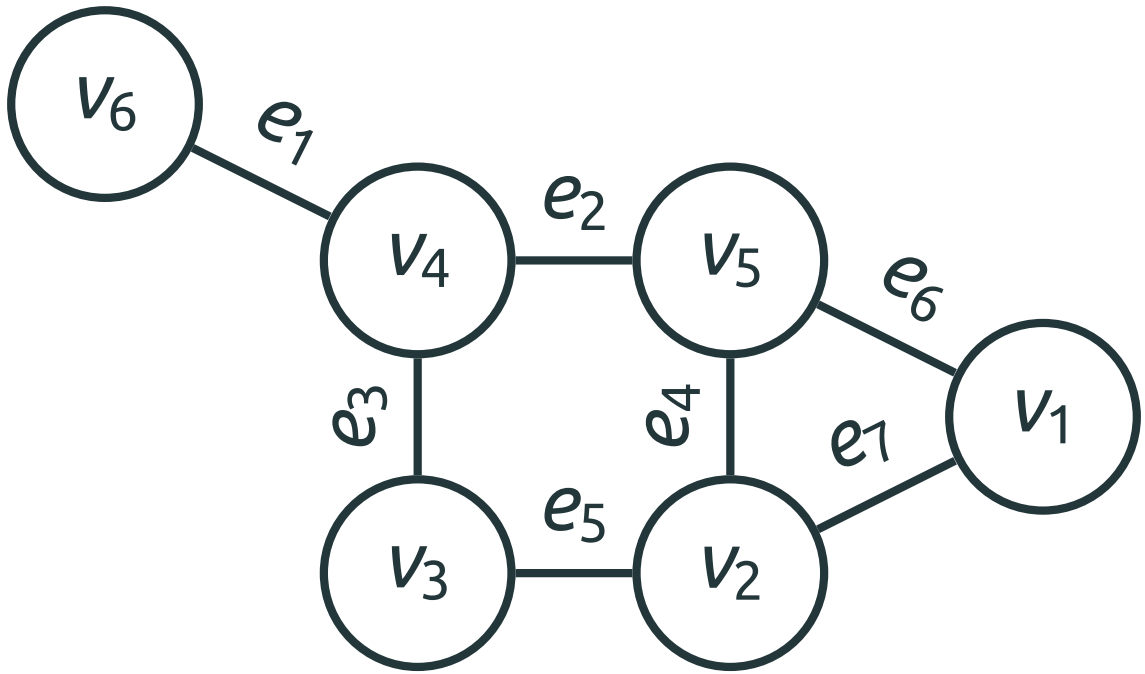
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Walks

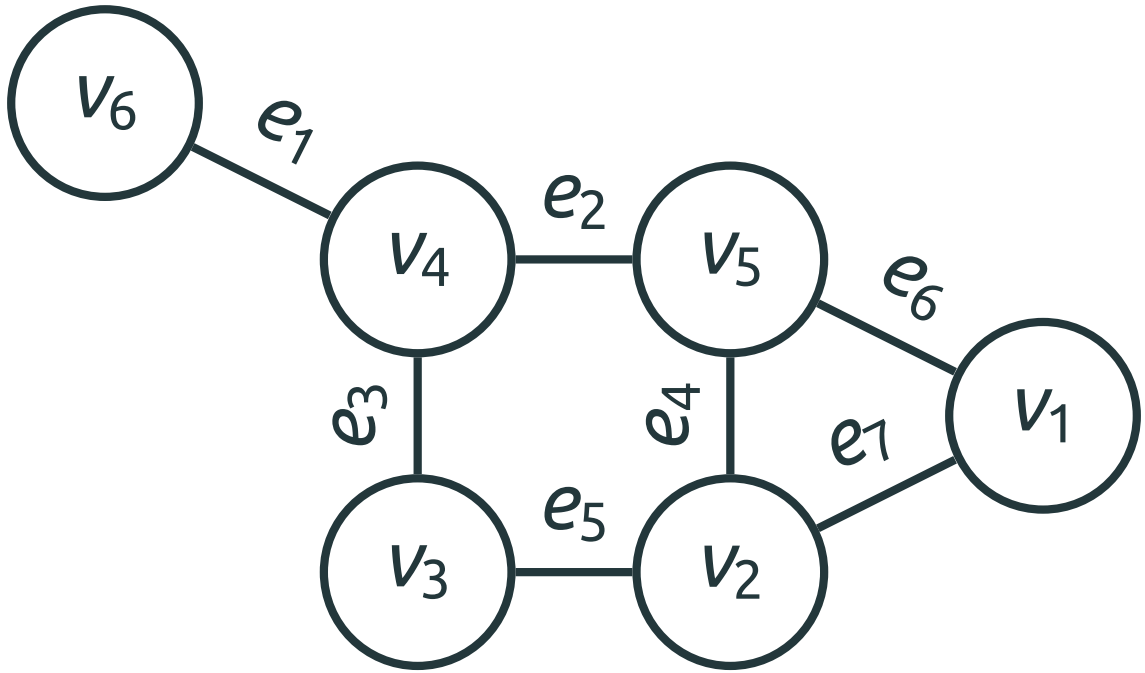
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- A Simple Path is a walk where all vertices are distinct

Walks: Examples



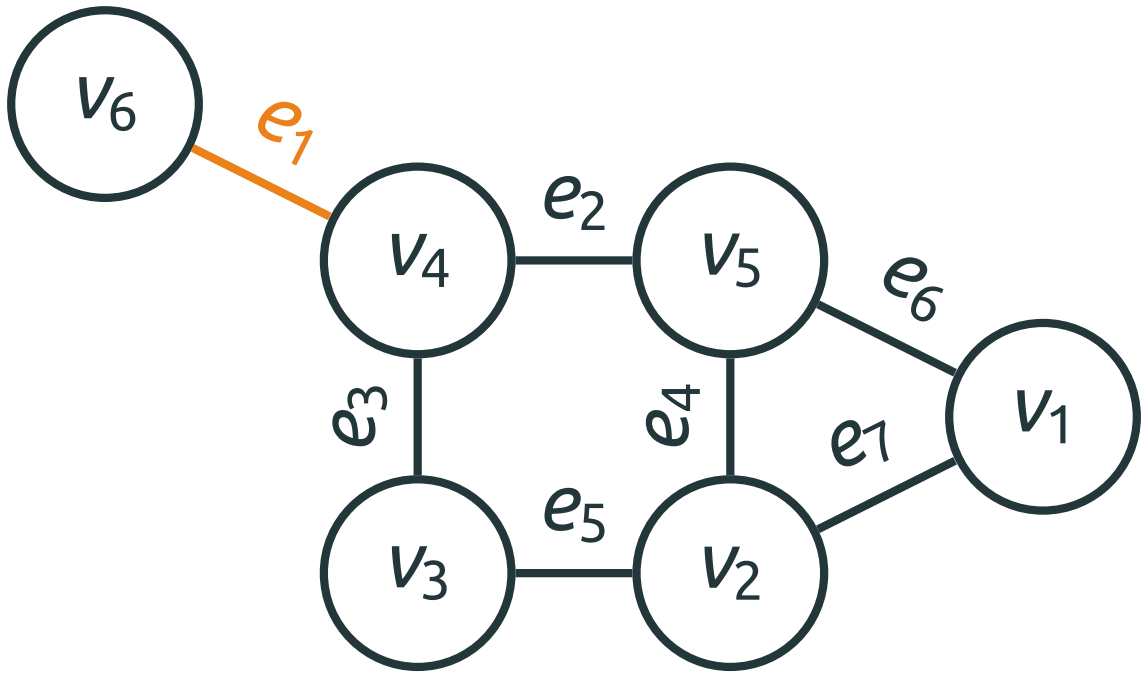
Walks: Examples

A walk of length 6: $(e_1, e_2, e_4, e_5, e_3, e_1)$



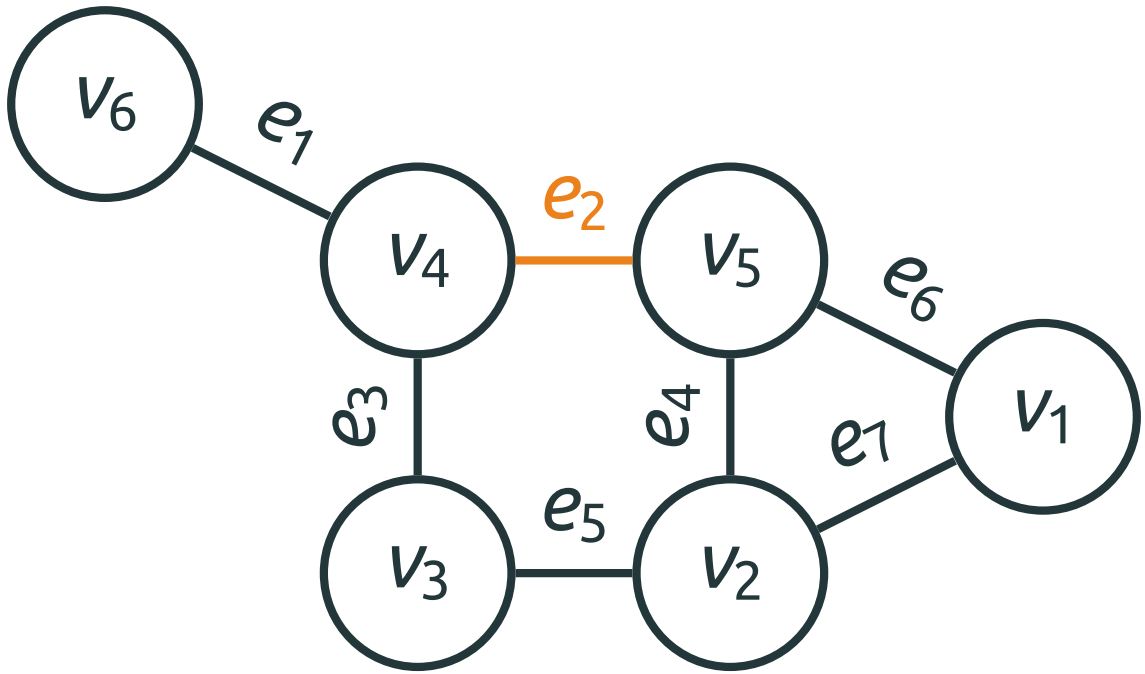
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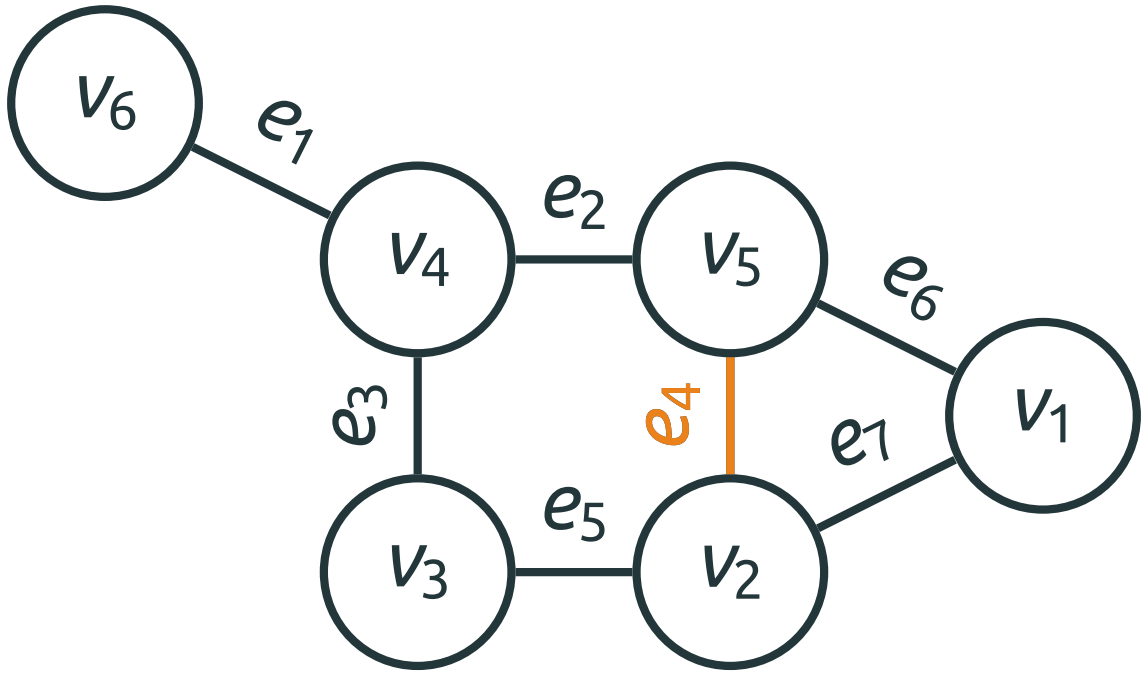
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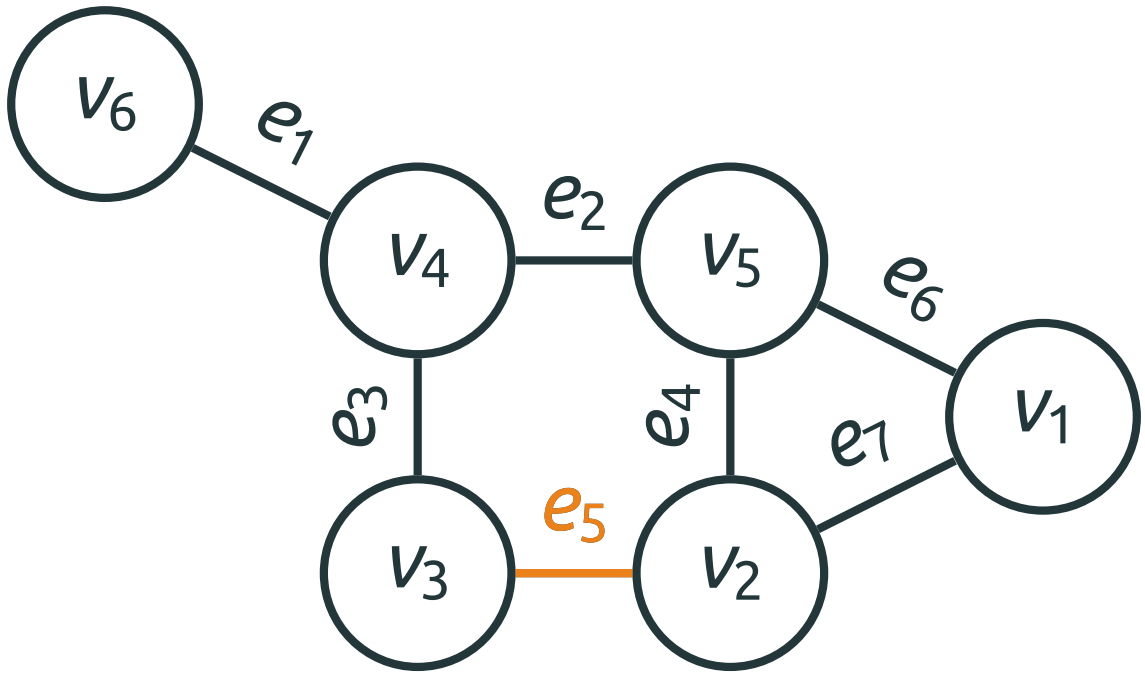
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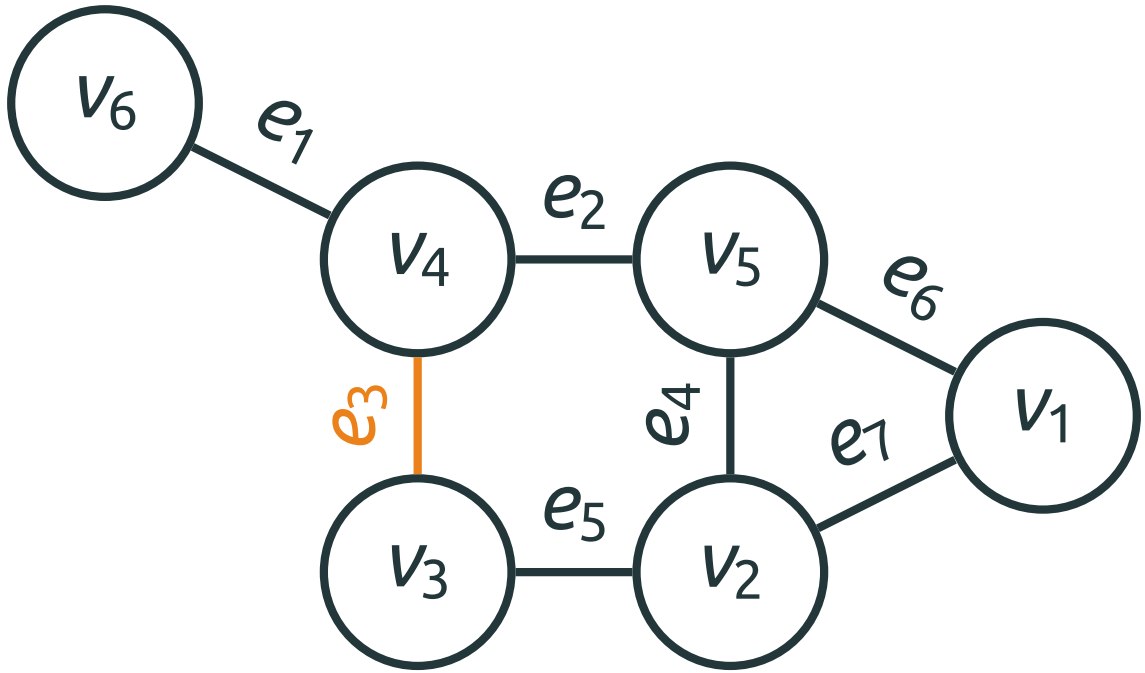
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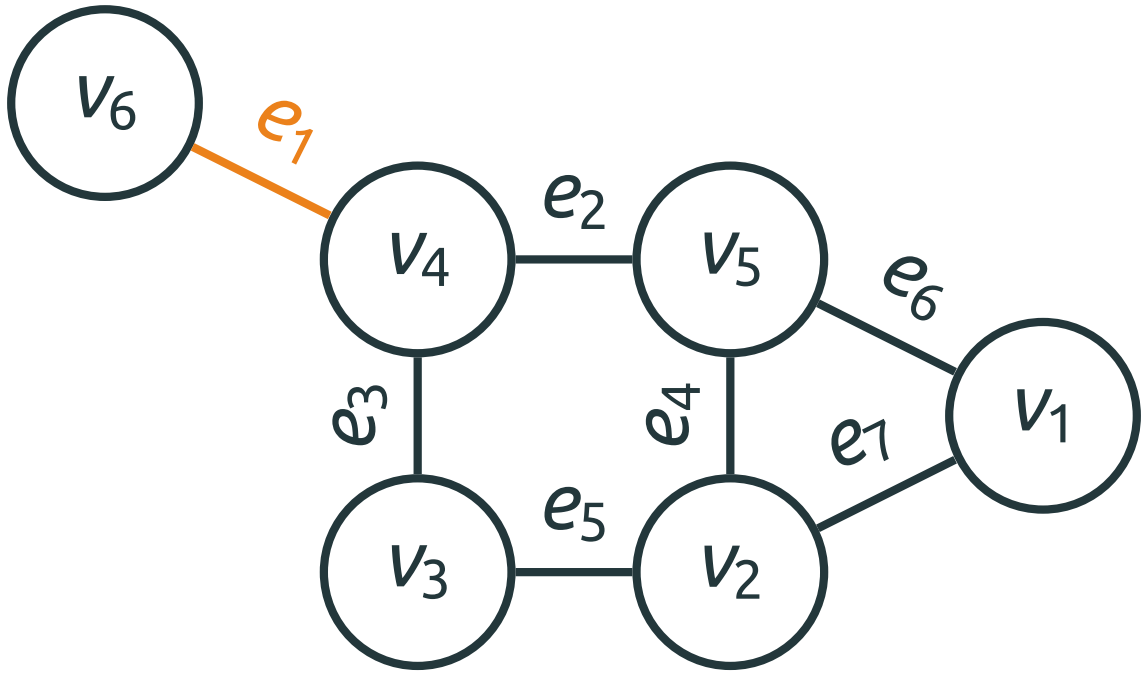
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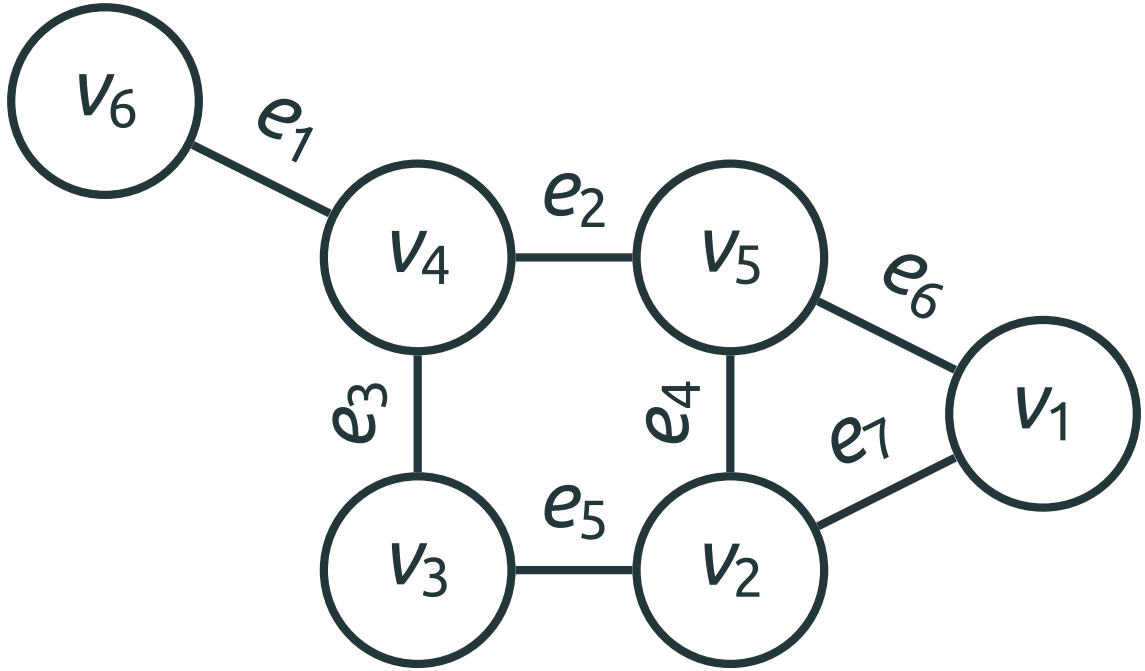
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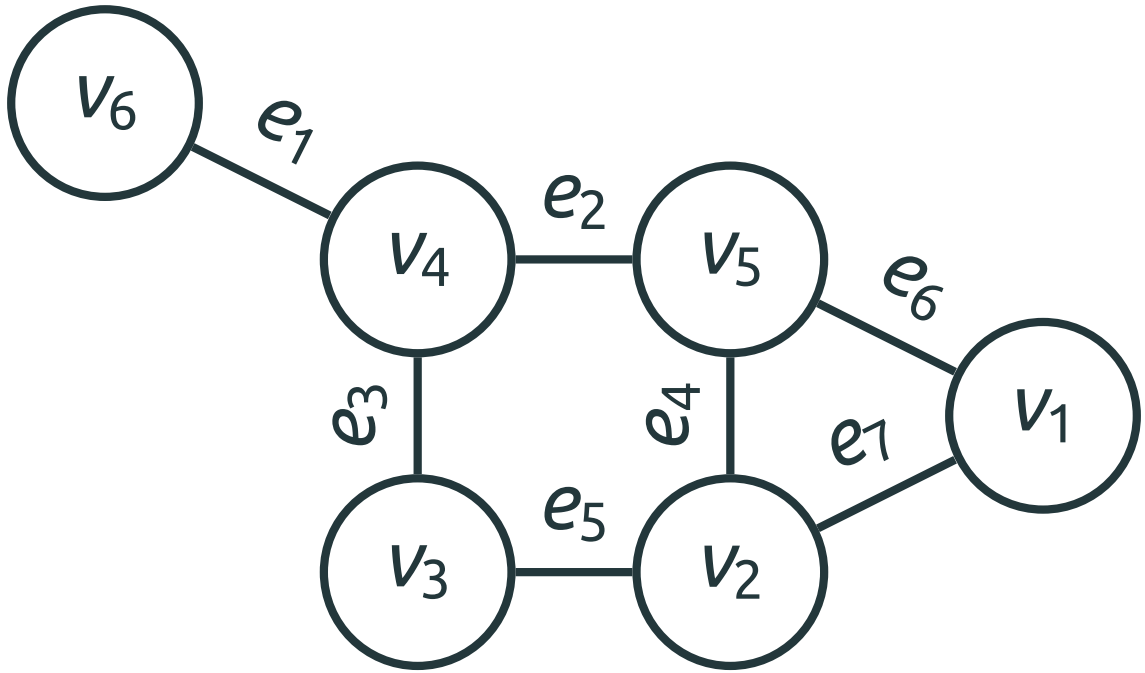
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Not a path: uses e_1 twice



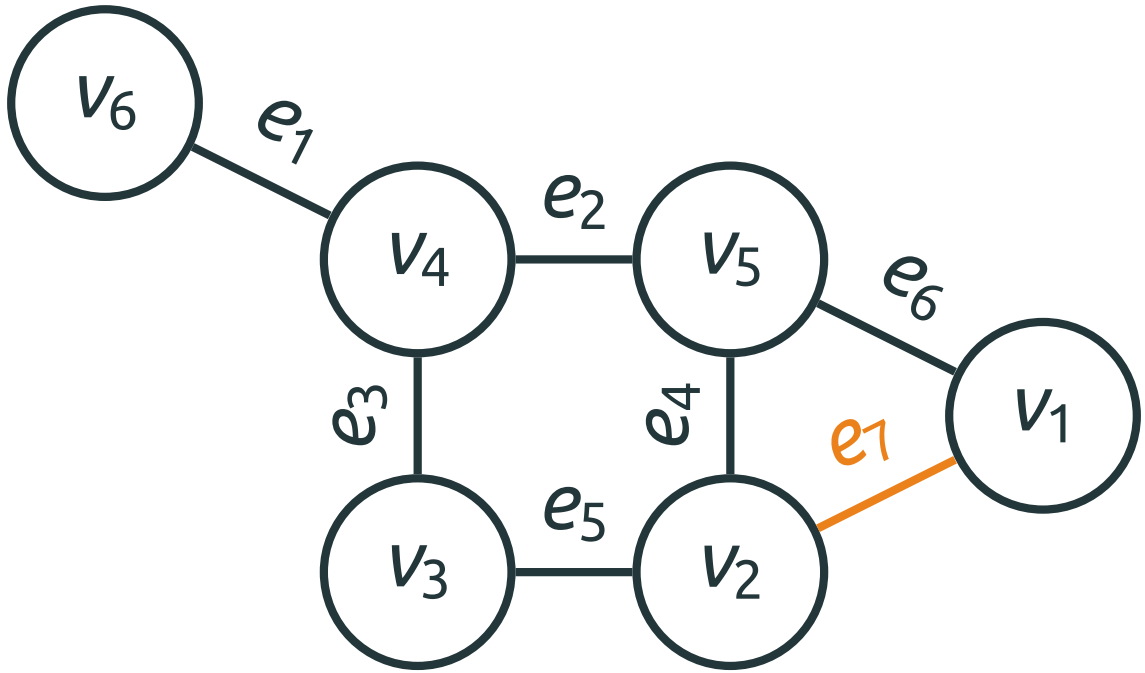
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A path of length 4: (e_7, e_6, e_4, e_5)



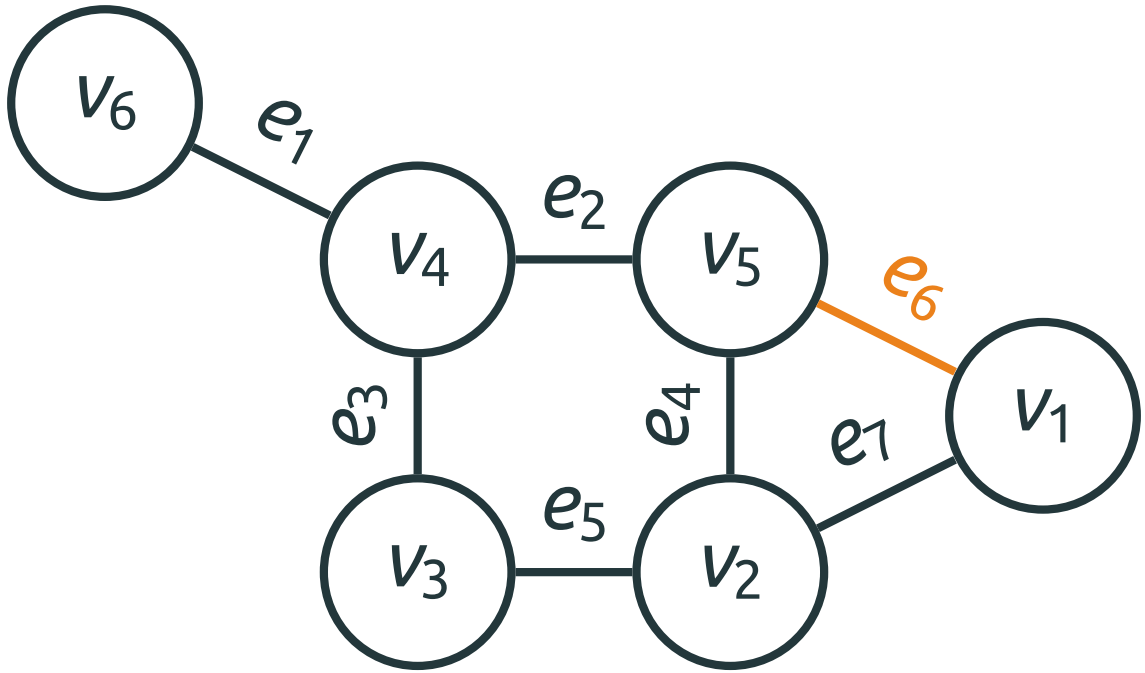
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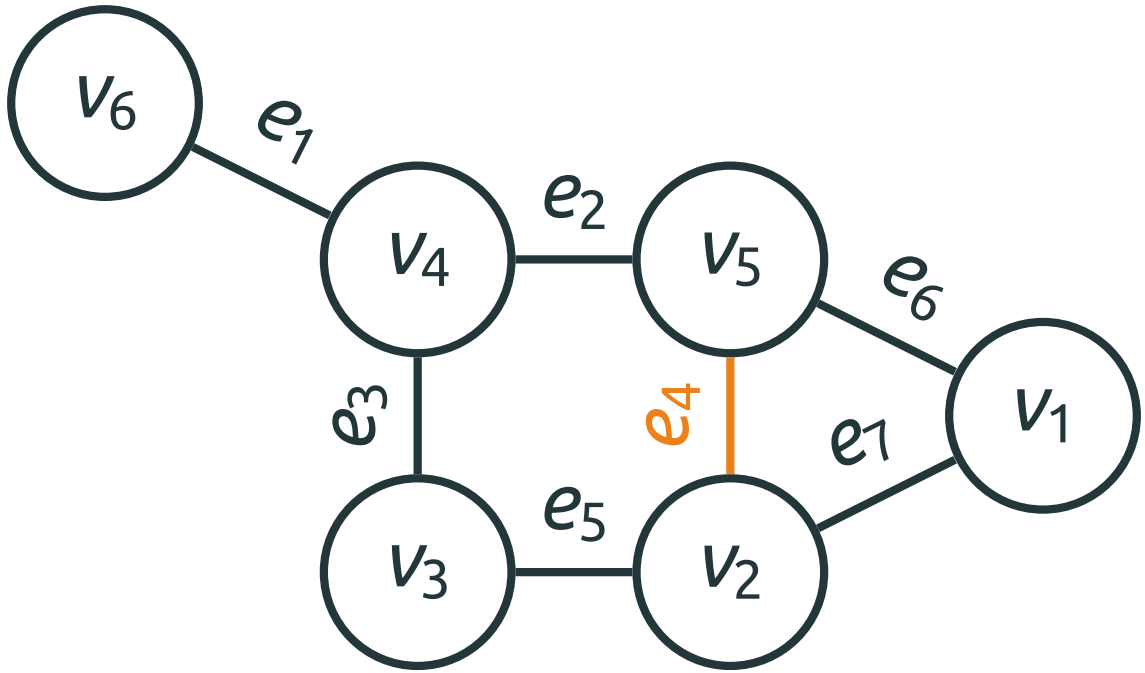
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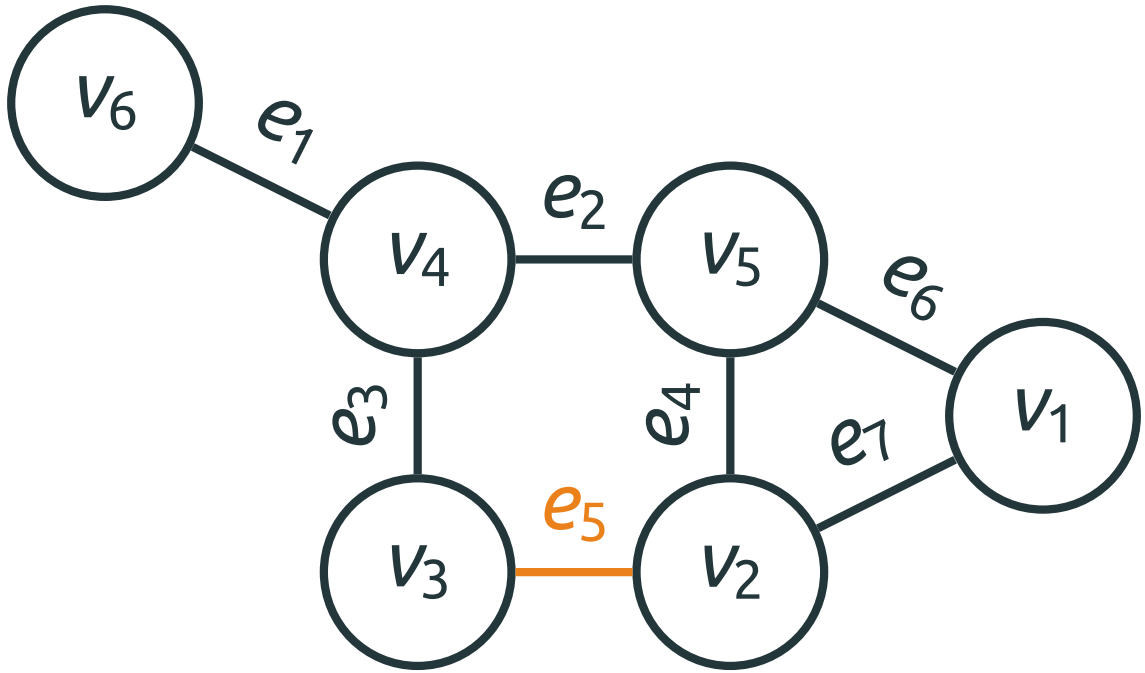
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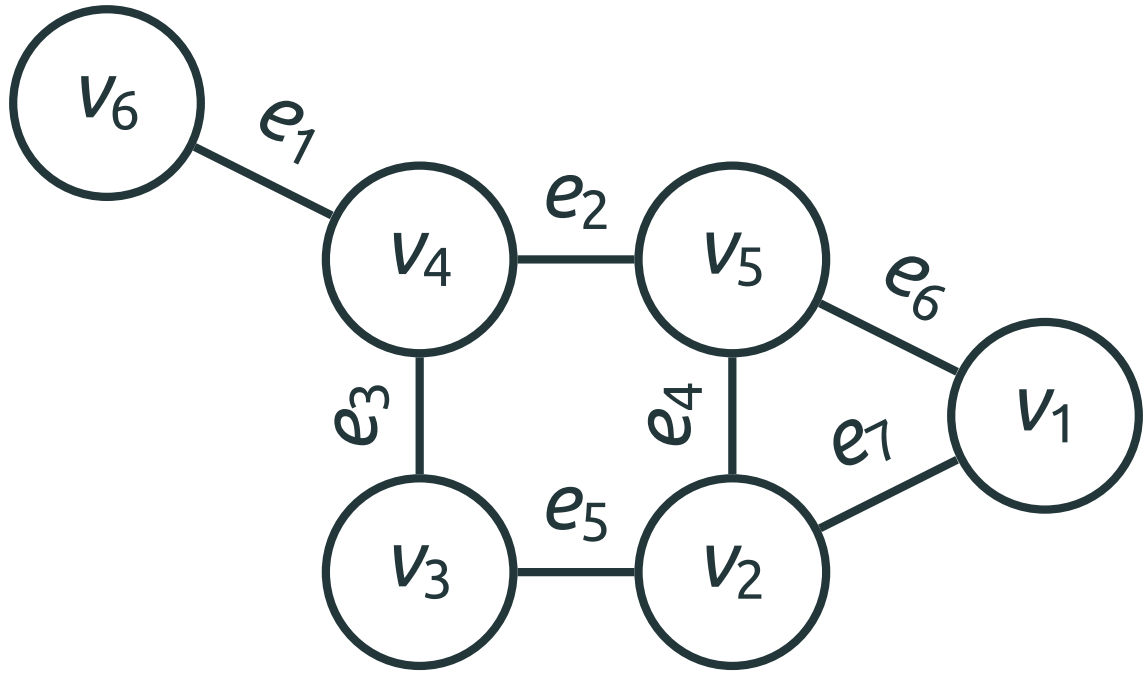
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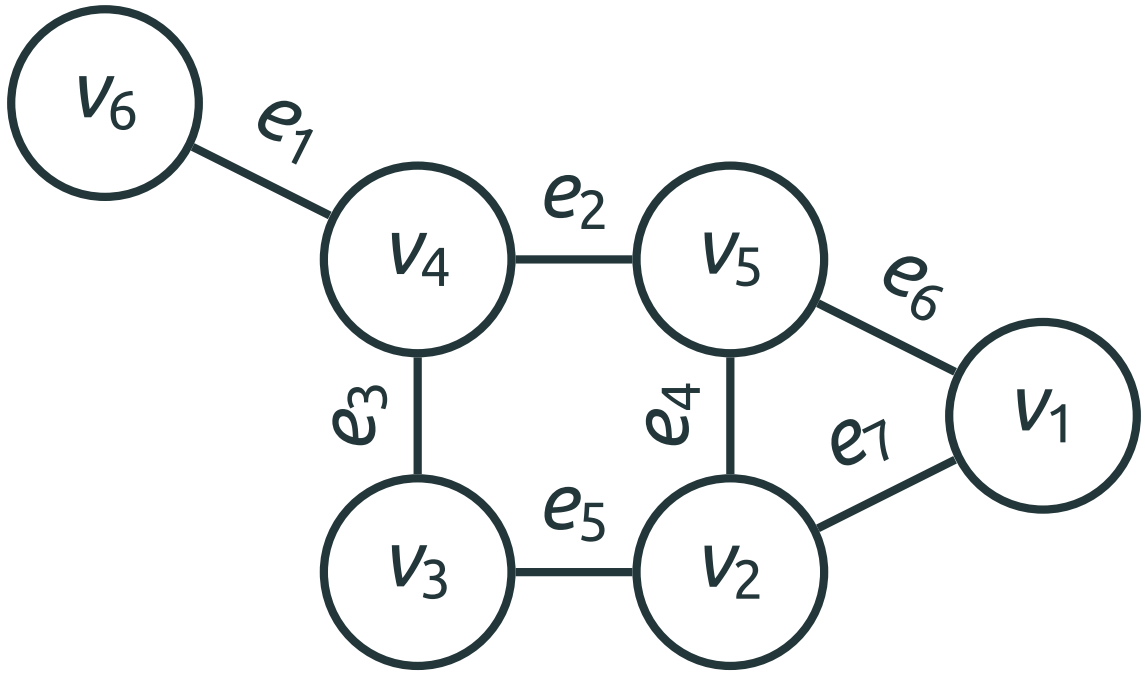
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A path of length 4: (e_7, e_6, e_4, e_5)
Not a simple path: visits v_2 twice



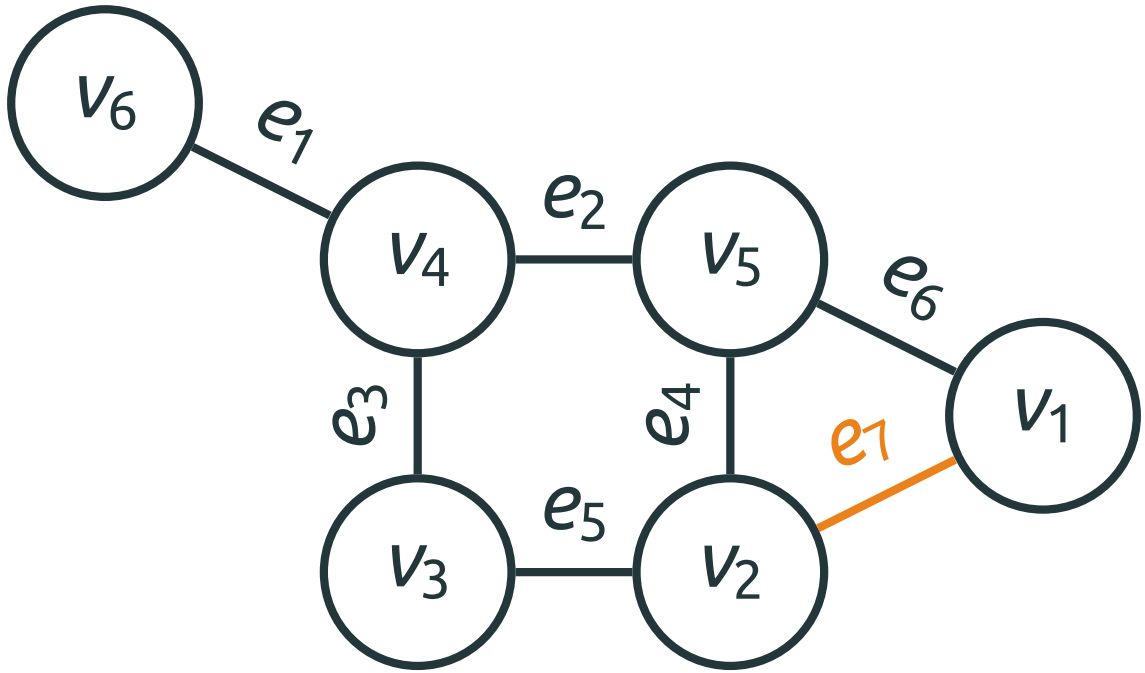
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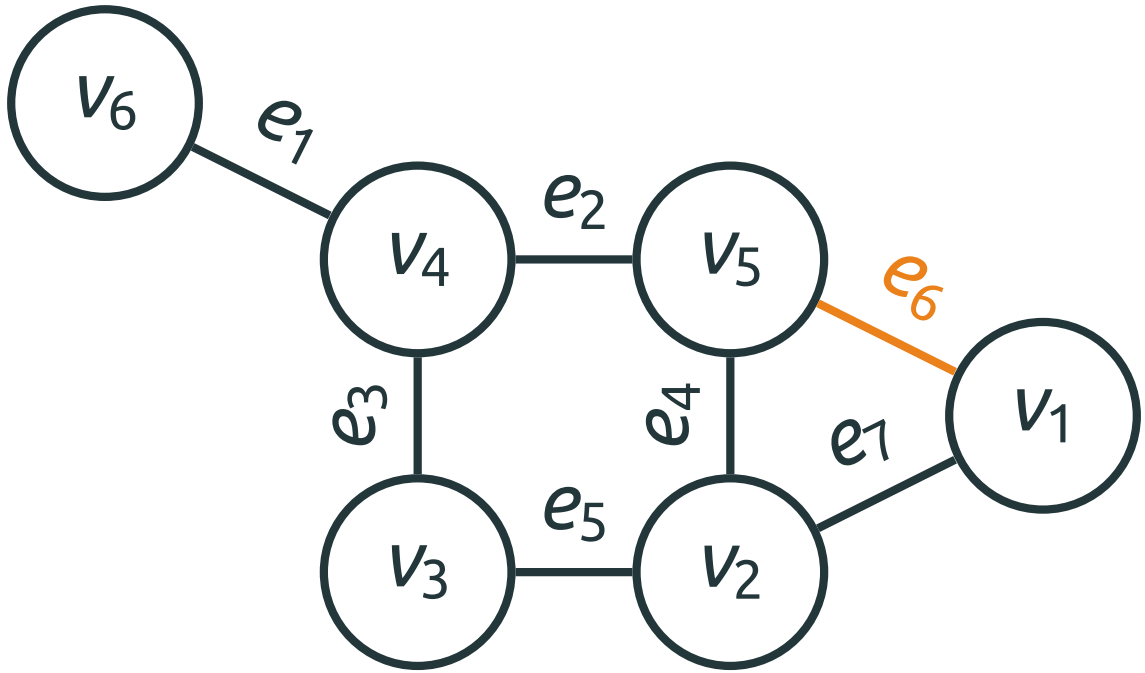
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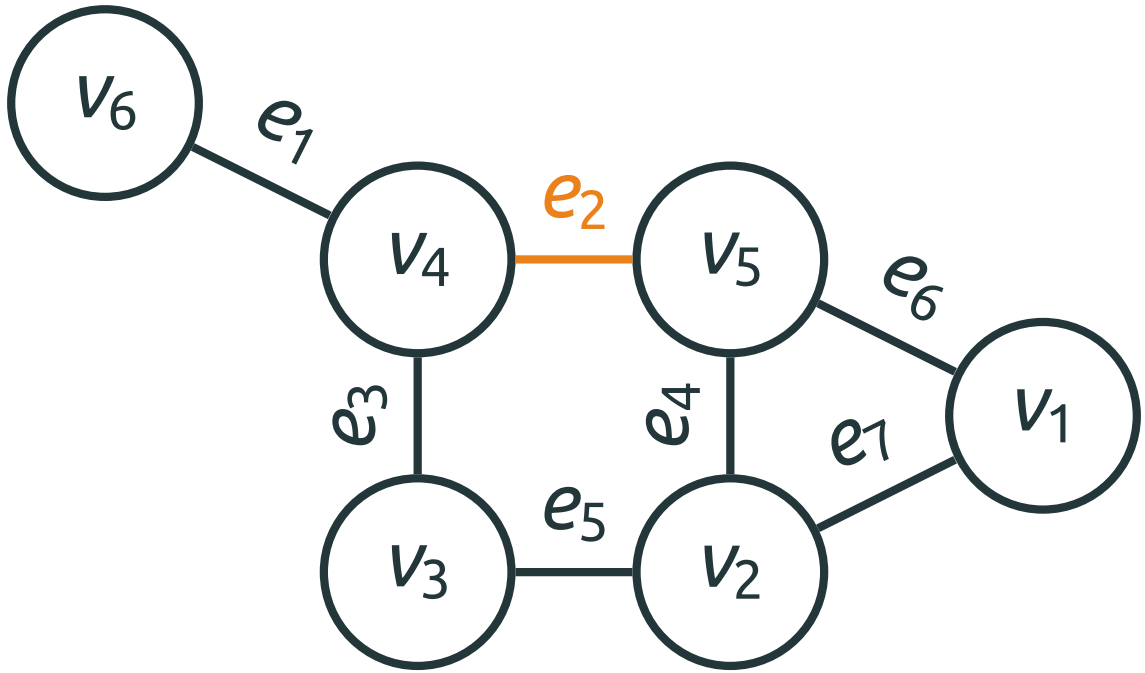
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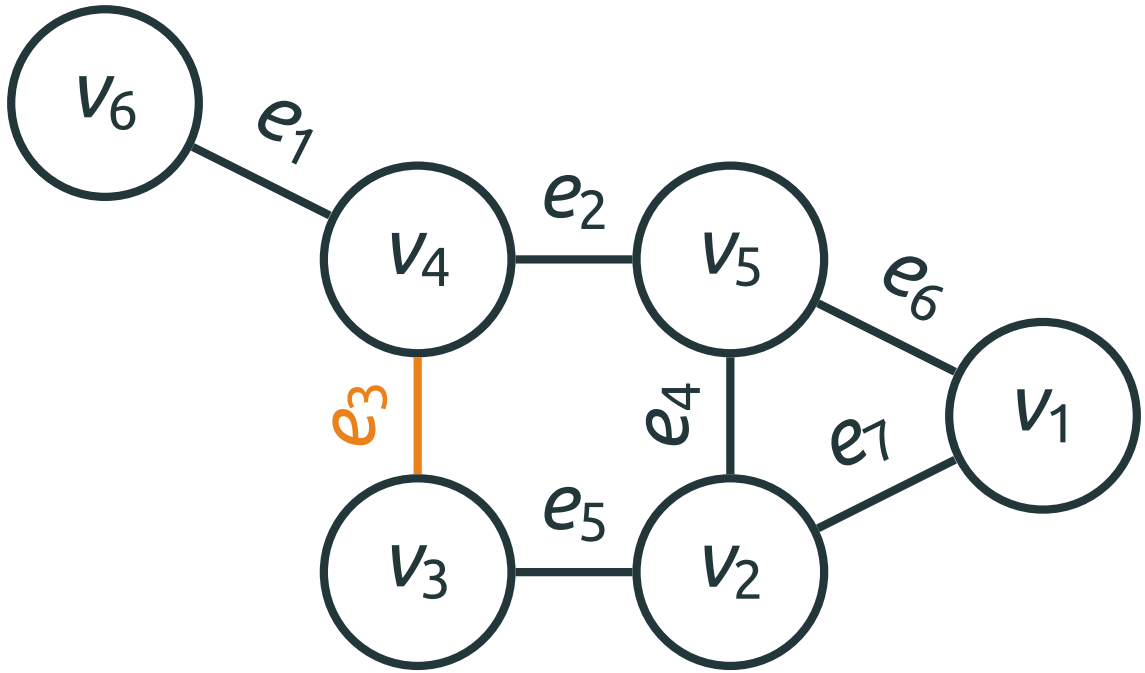
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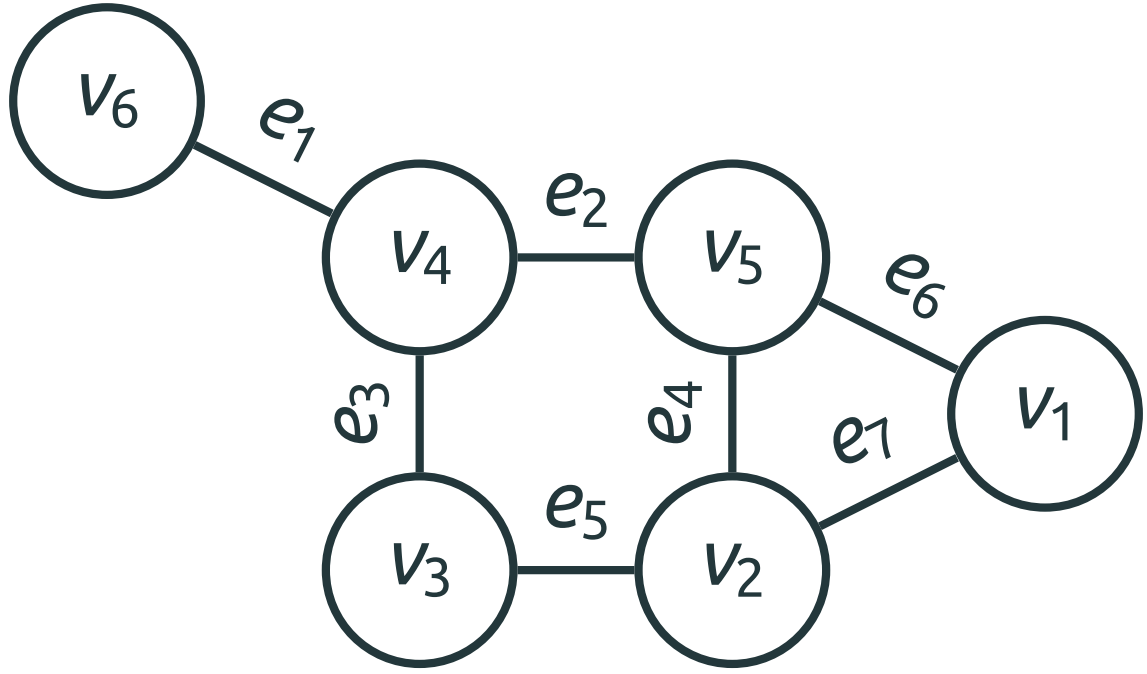
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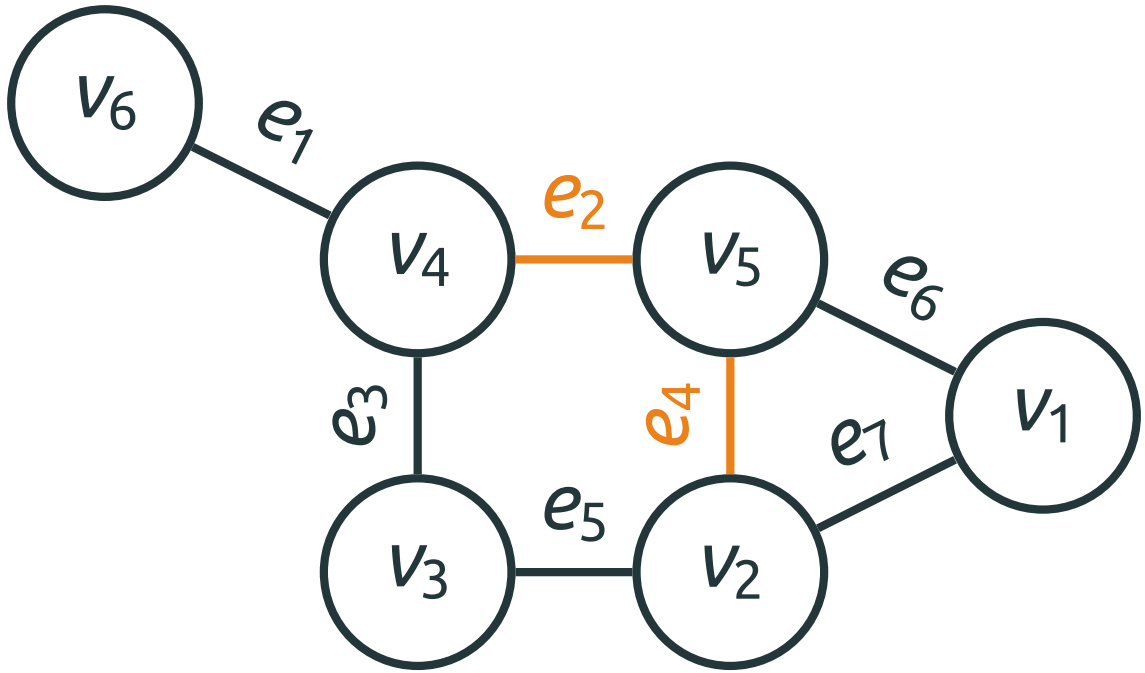
Walks: Examples

It is sometimes convenient to specify a path (walk) by a list of its vertices



Walks: Examples

(v_4, v_5, v_2) is a path of length 2



Cycles

- A Cycle in a graph is a path whose first vertex is the same as the last one

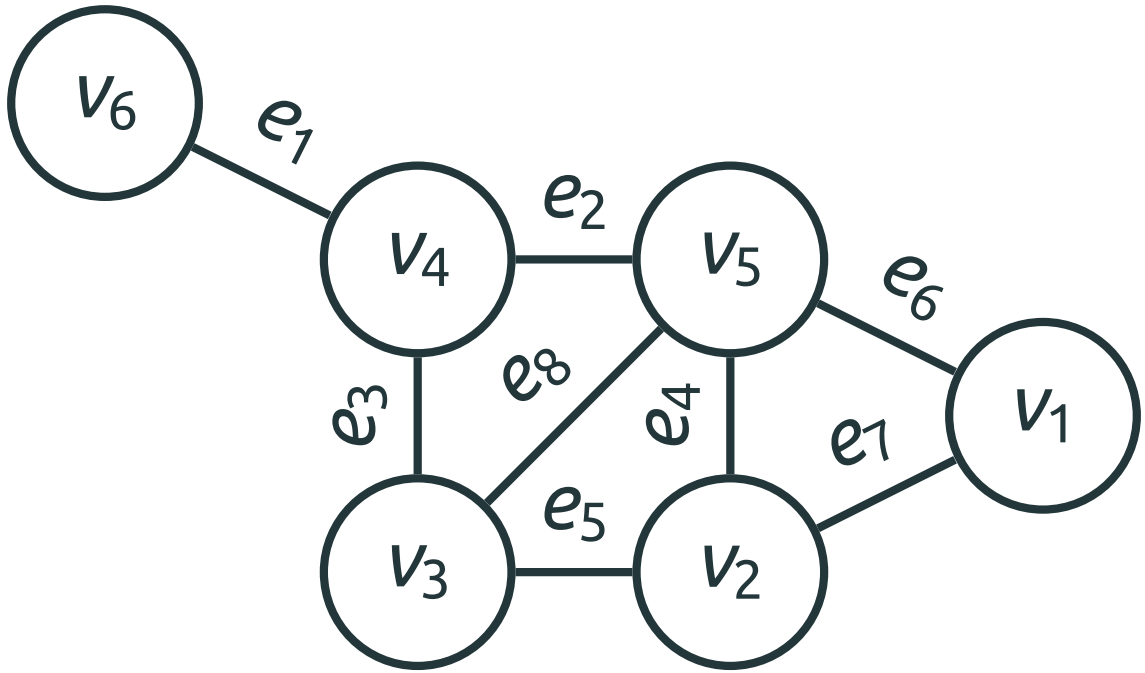
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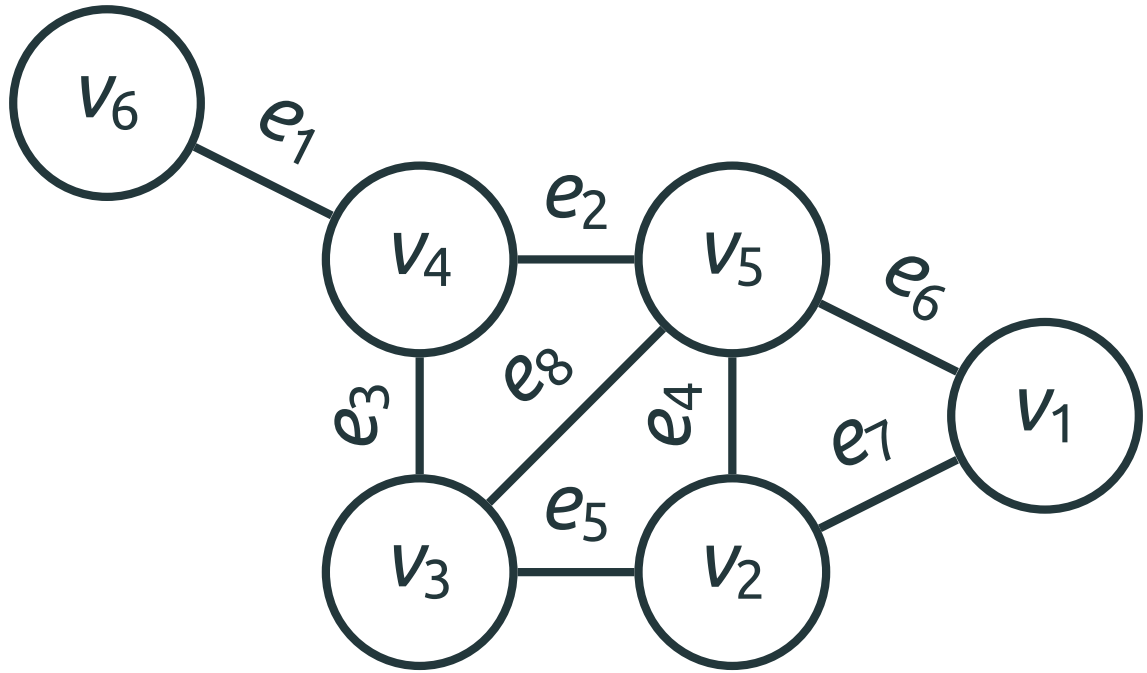
- A Cycle in a graph is a path whose first vertex is the same as the last one
- In particular, all the edges in a Cycle are distinct
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Cycles: Examples



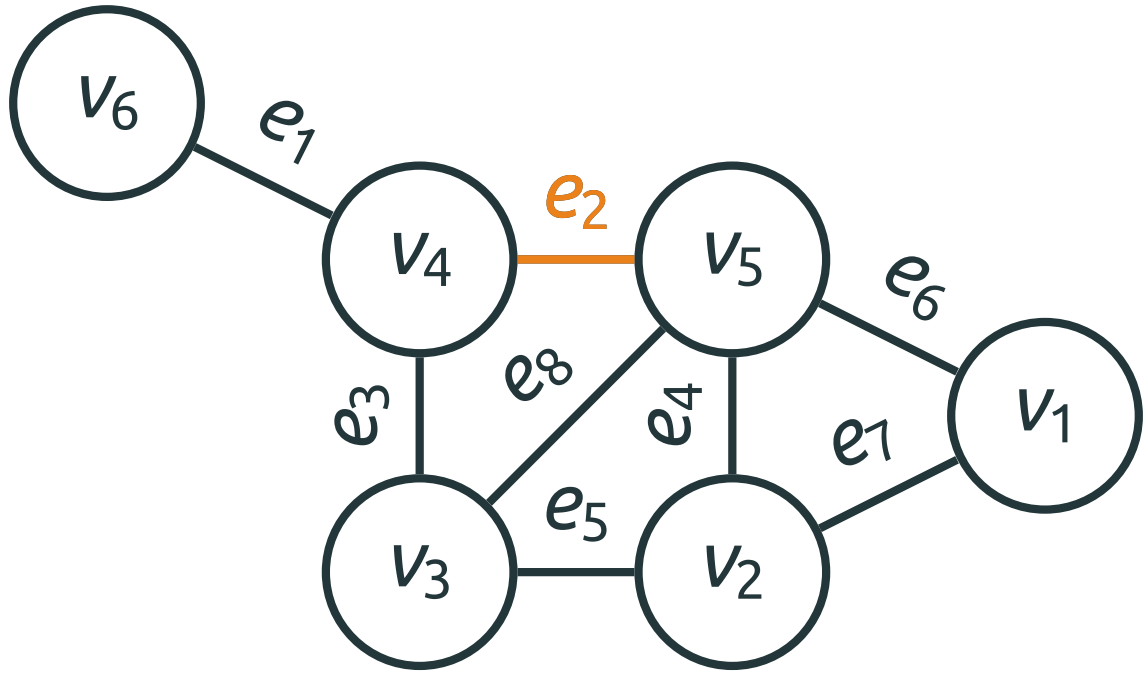
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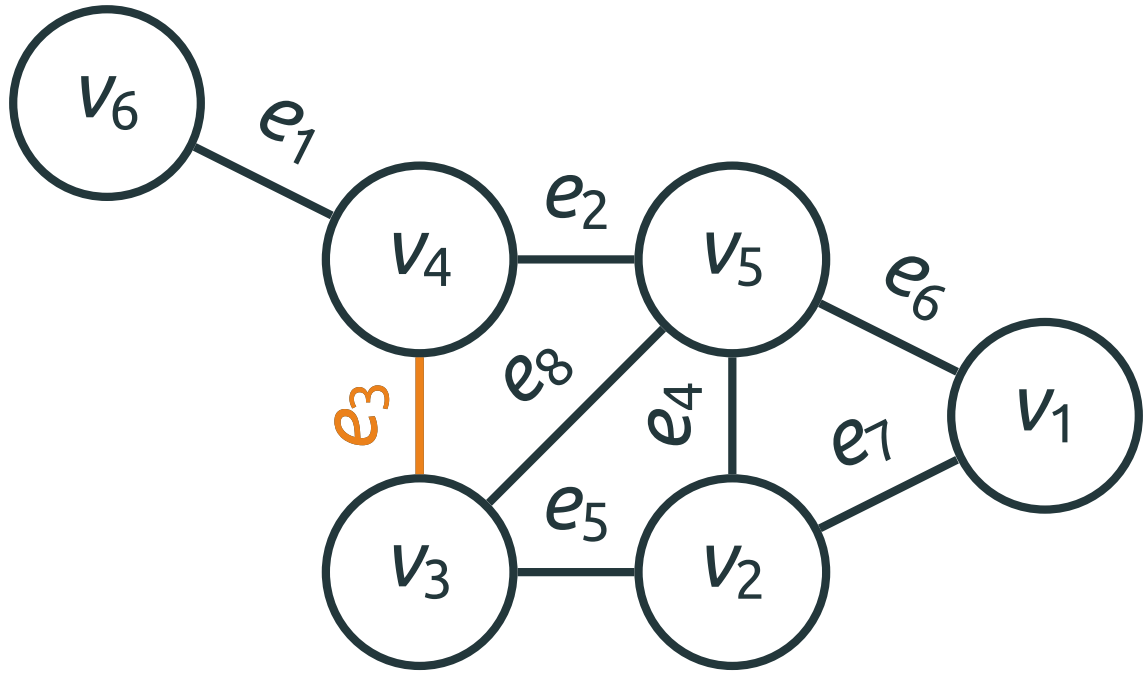
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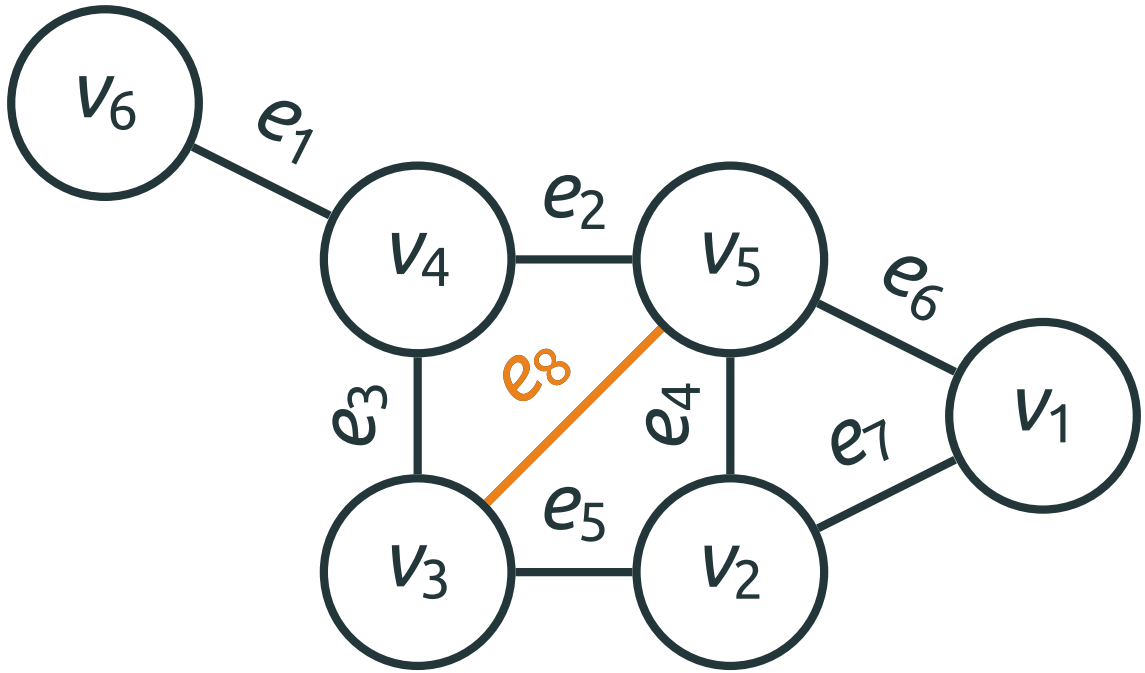
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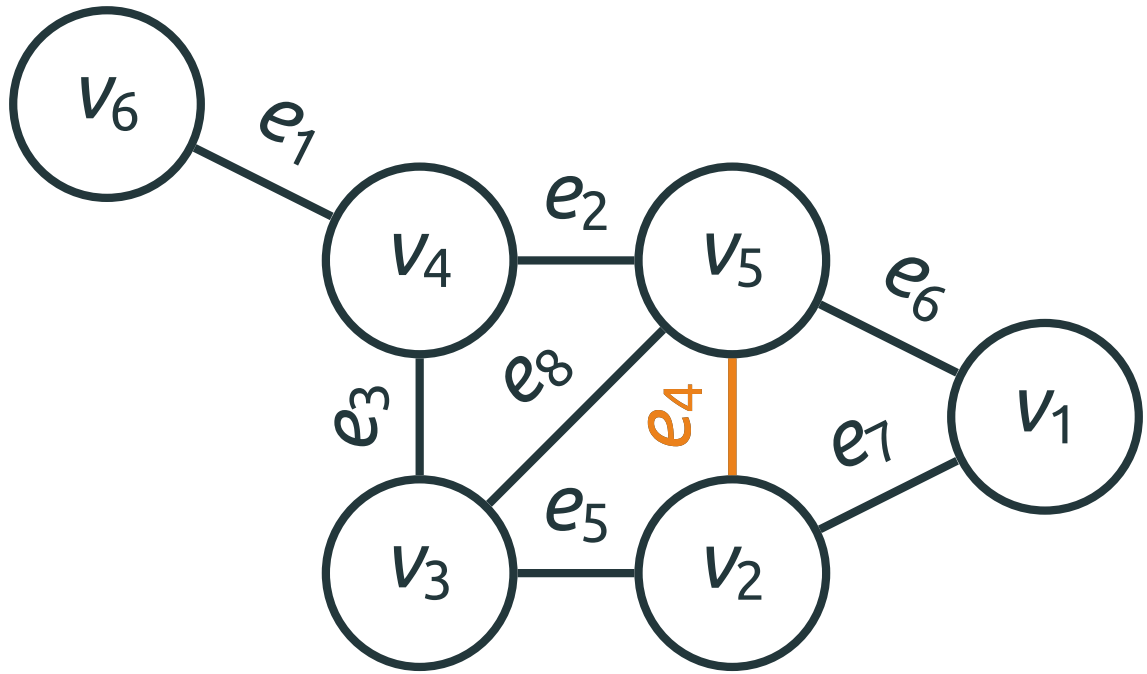
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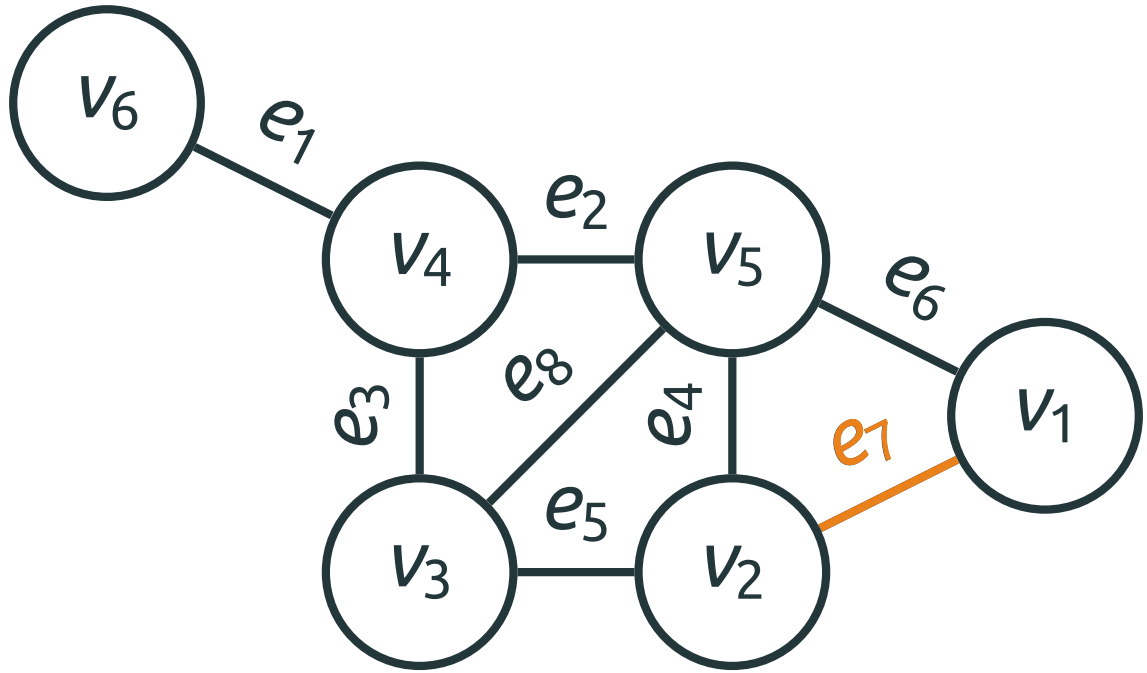
Cycles: Examples

A cycle of length 6: $(e_2, e_3, e_8, e_4, e_7, e_6)$



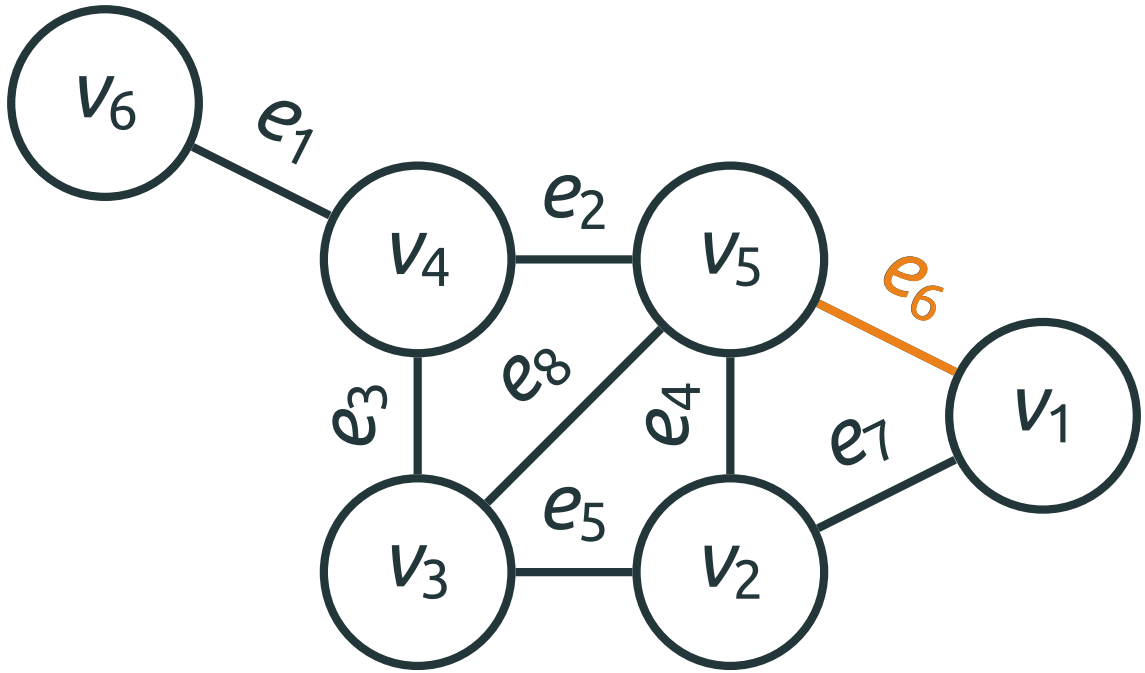
Cycles: Examples

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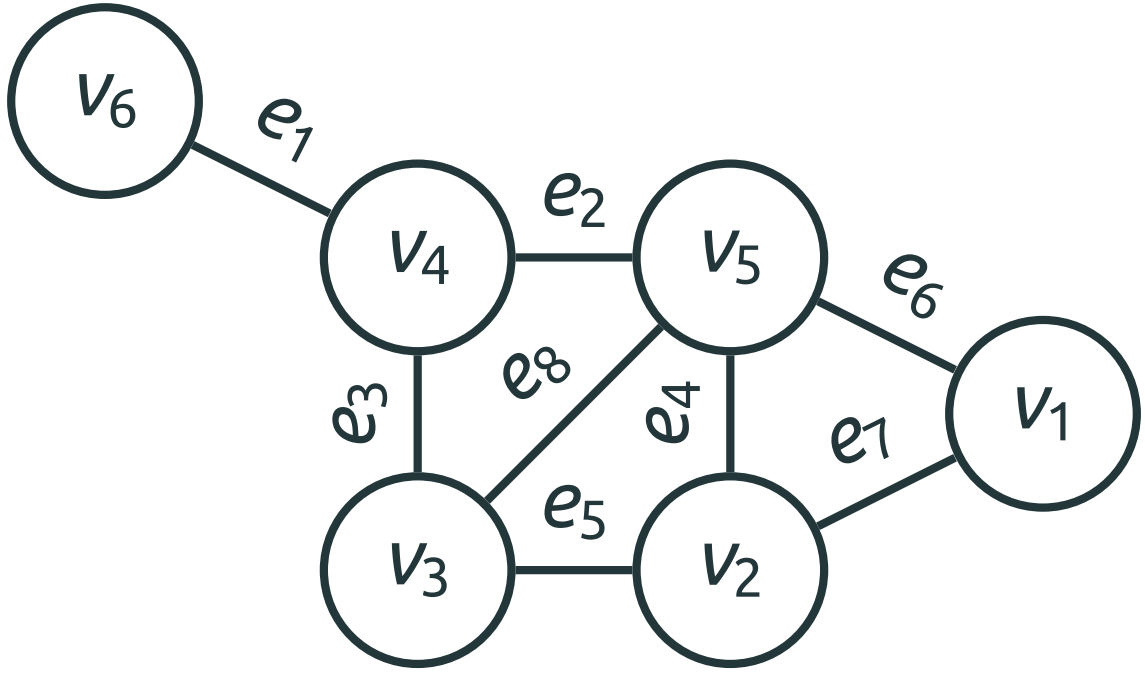
Cycles: Examples

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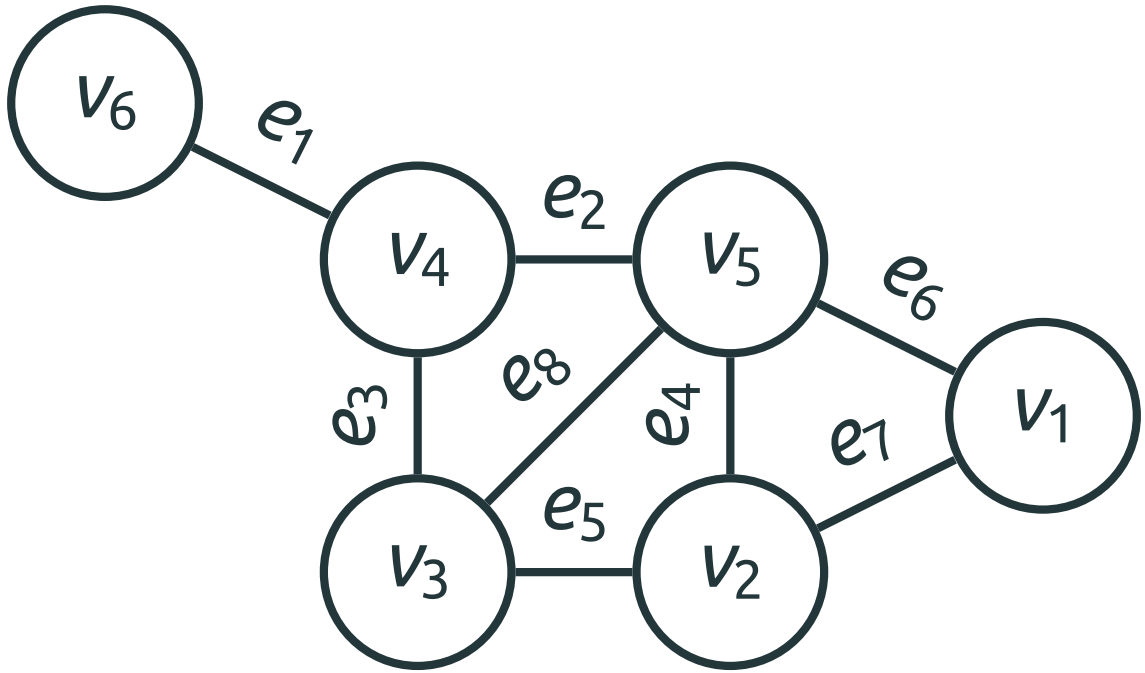
Cycles: Examples

A cycle of length 6: $(e_2, e_3, e_8, e_4, e_7, e_6)$
Not a simple cycle: visits v_5 three times



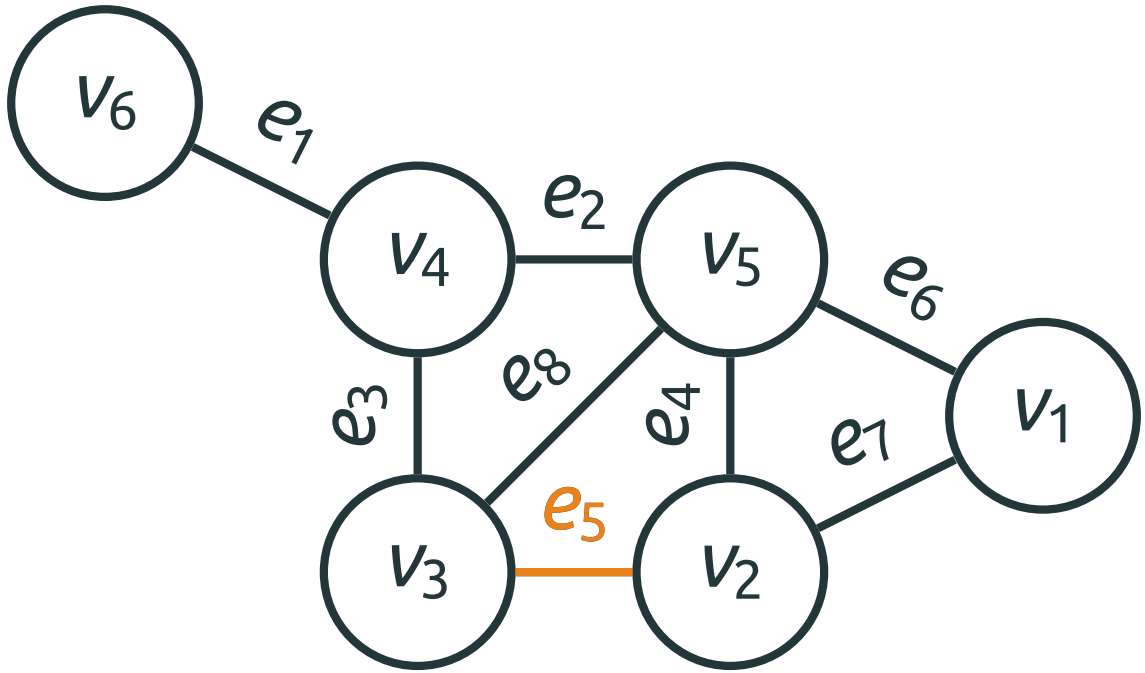
Cycles: Examples

A simple cycle of length 4: (e_5, e_4, e_2, e_3)



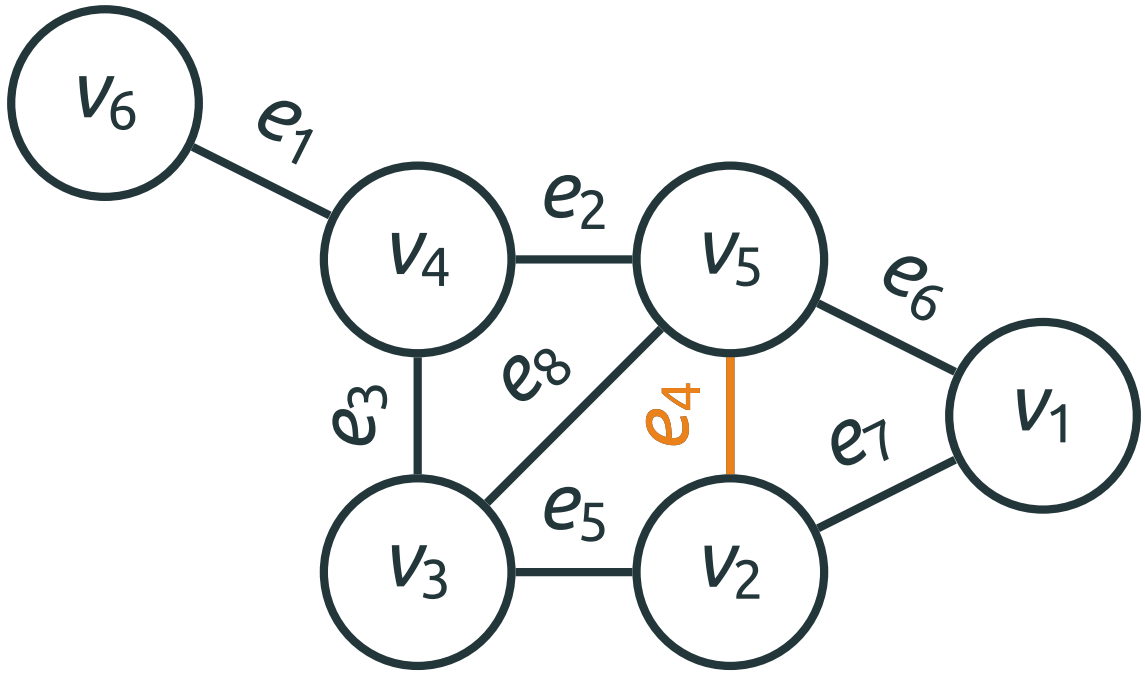
Cycles: Examples

A simple cycle of length 4: (e_5, e_4, e_2, e_3)



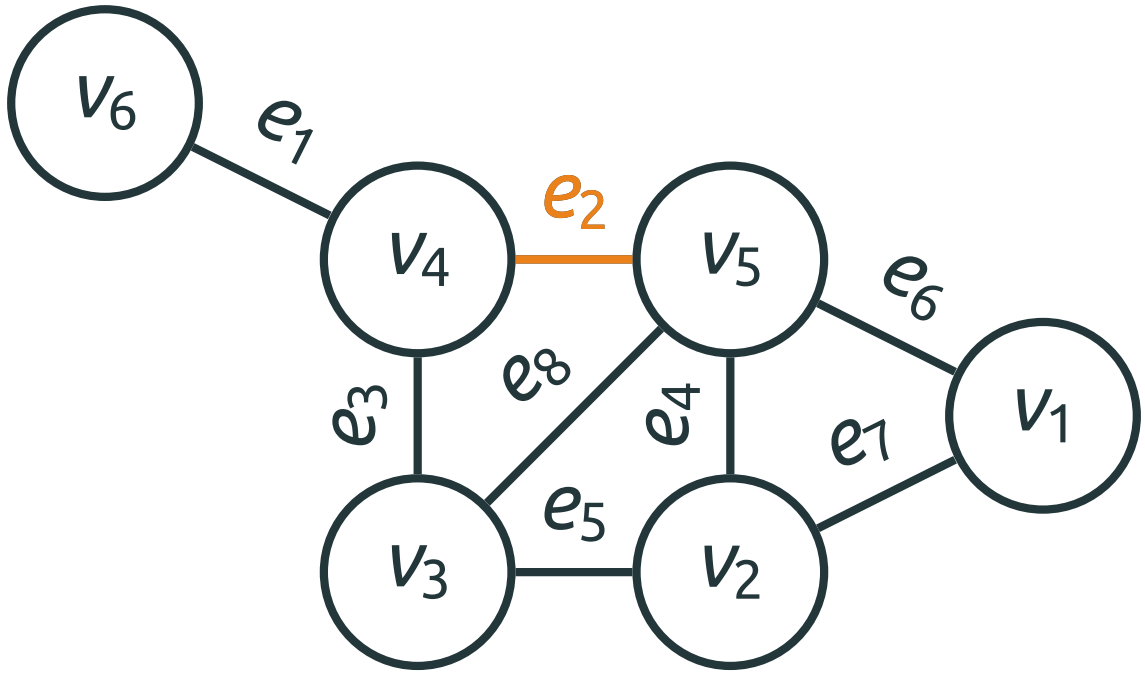
Cycles: Examples

A simple cycle of length 4: (e_5, e_4, e_2, e_3)



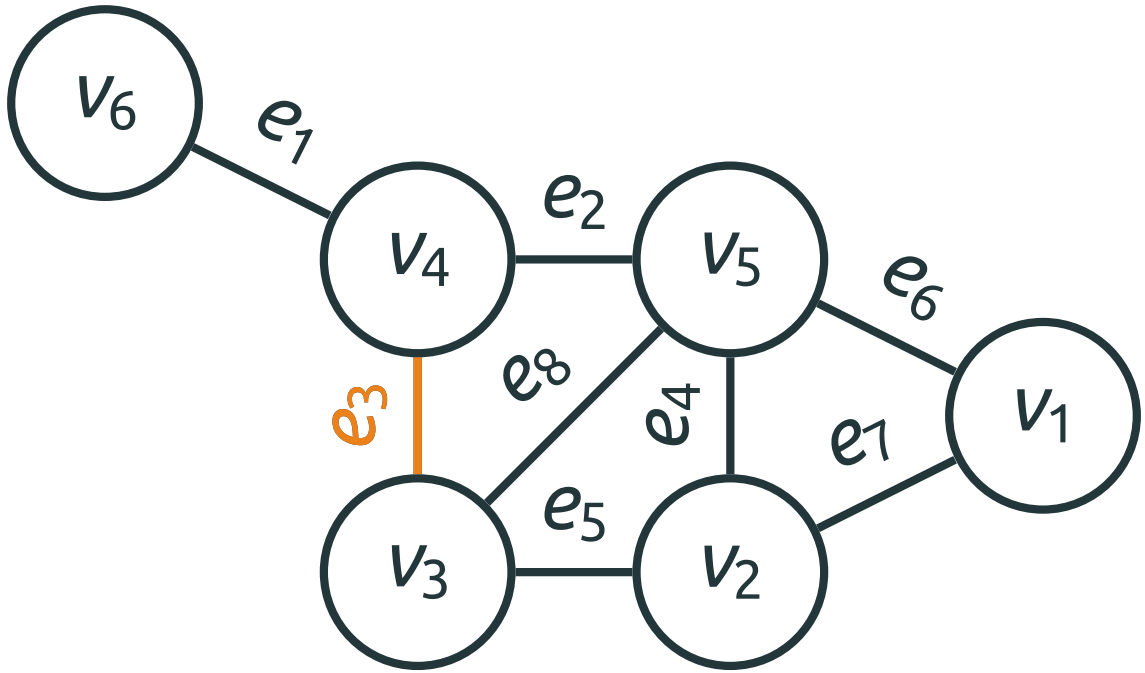
Cycles: Examples

A simple cycle of length 4: (e_5, e_4, e_2, e_3)



Cycles: Examples

A simple cycle of length 4: (e_5, e_4, e_2, e_3)



Outline

The Degree of a Vertex

Paths

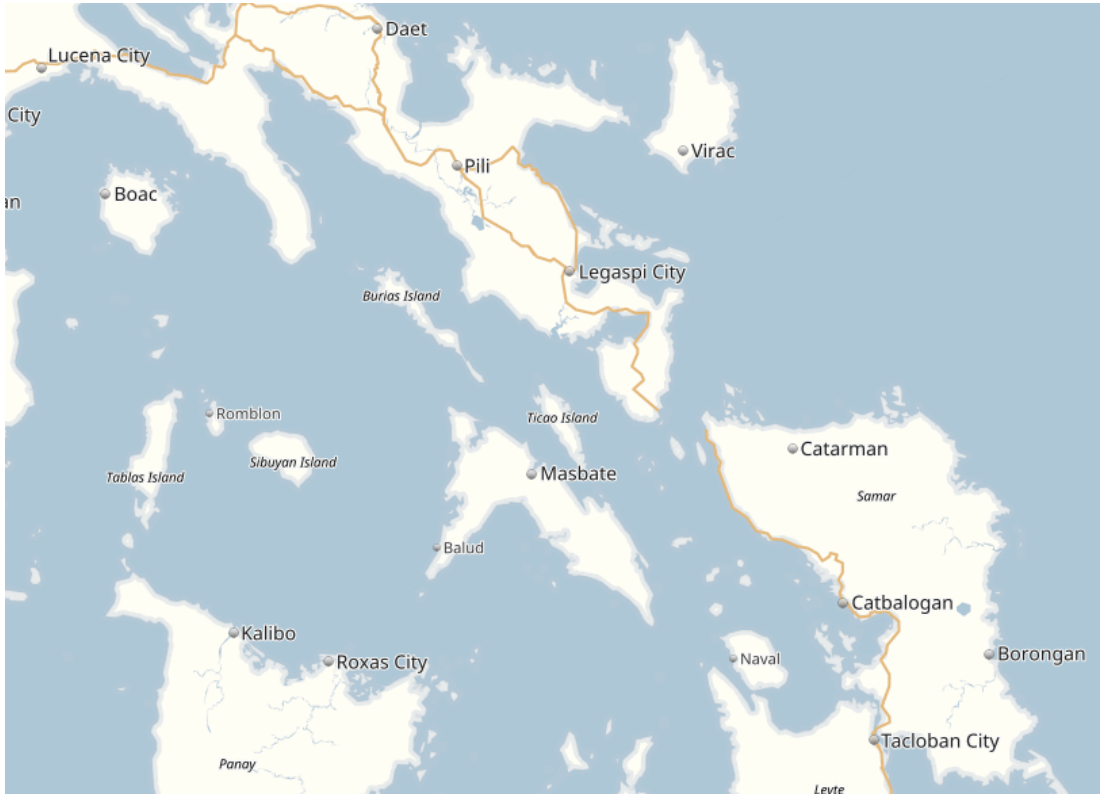
Connectivity

Directed Graphs

Weighted Graphs

Connected Components

The number of islands



Connectivity

- A graph is called Connected if there is a path between every pair of its vertices

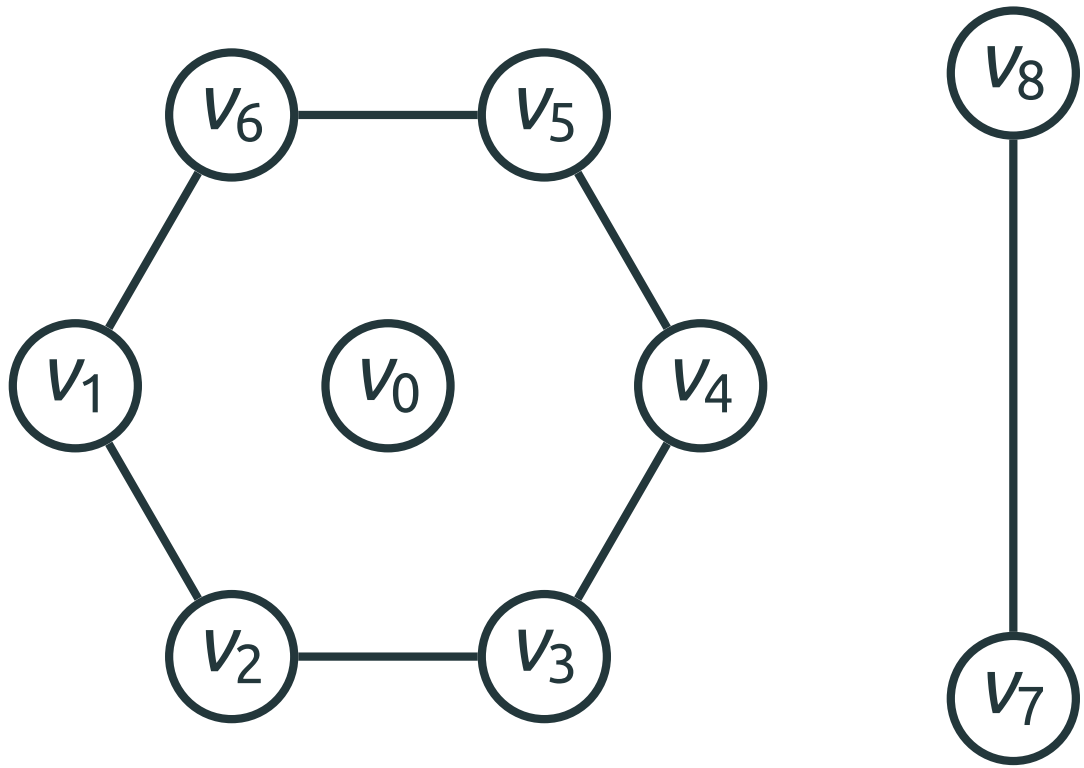
Connectivity

- A graph is called Connected if there is a path between every pair of its vertices
- A Connected Component of a graph G is a maximal connected subgraph of G

Connectivity

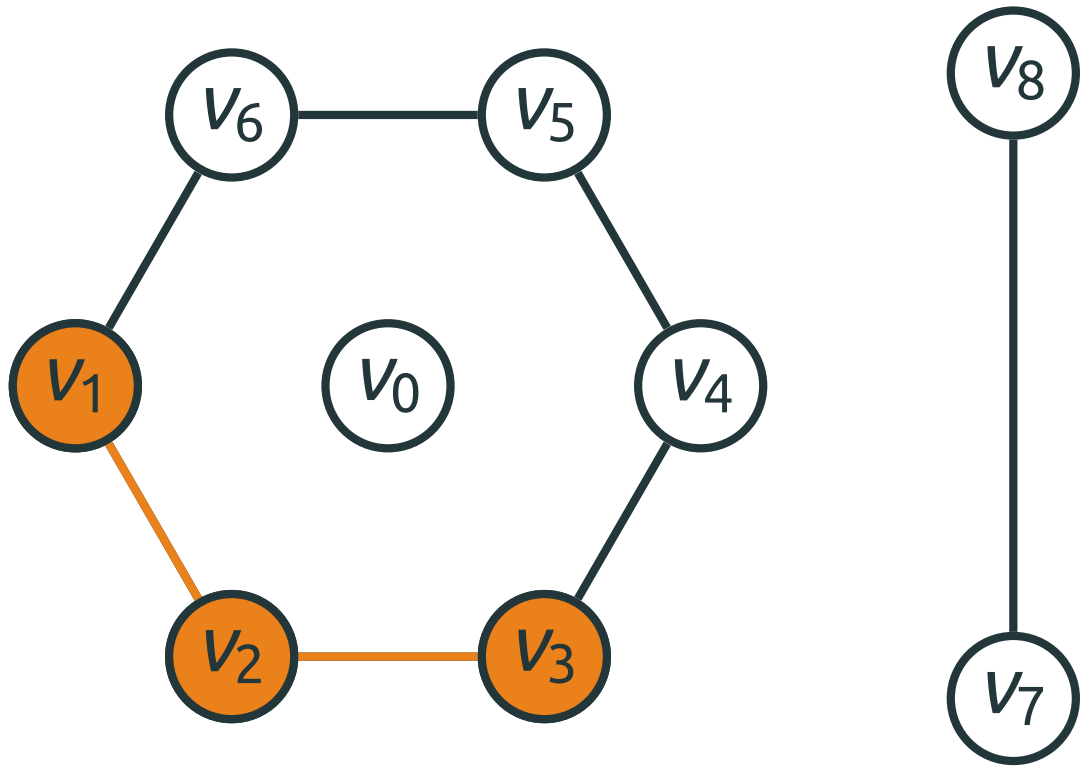
- A graph is called Connected if there is a path between every pair of its vertices
- A Connected Component of a graph G is a maximal connected subgraph of G
- I.e., a connected subgraph of G which is not contained in a larger connected subgraph of G

Connected Components: Examples



Connected Components: Examples

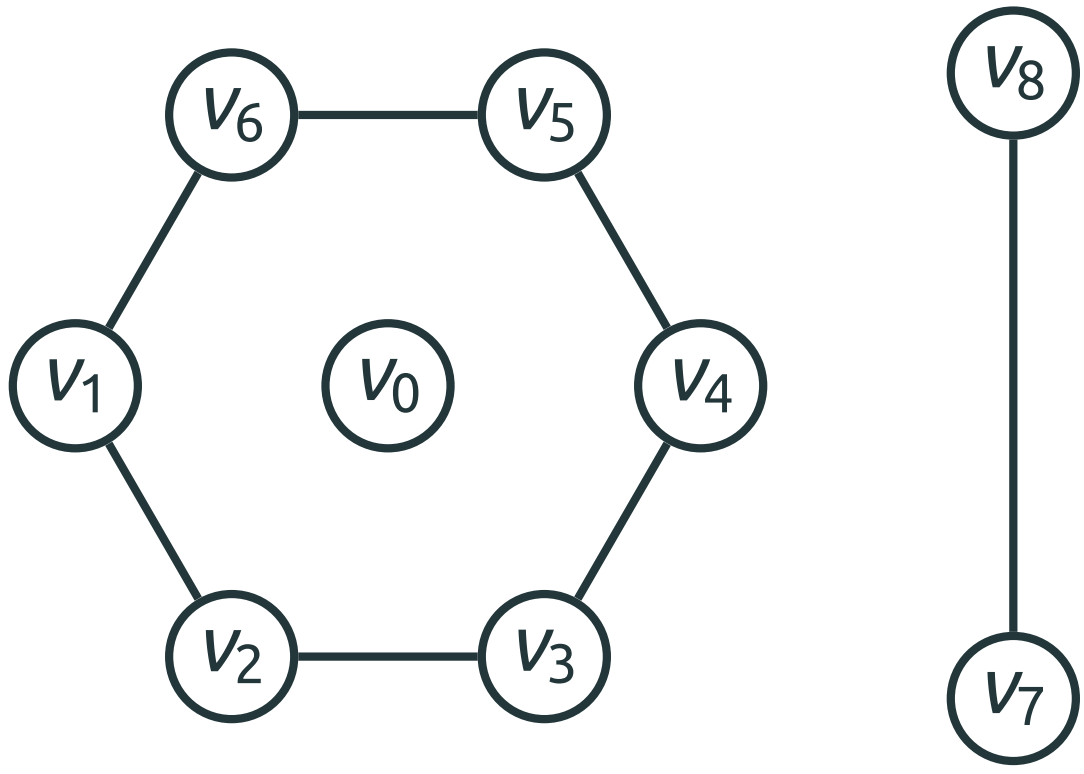
v_1, v_2, v_3 form a connected subgraph



Connected Components: Examples

v_1, v_2, v_3 form a connected subgraph

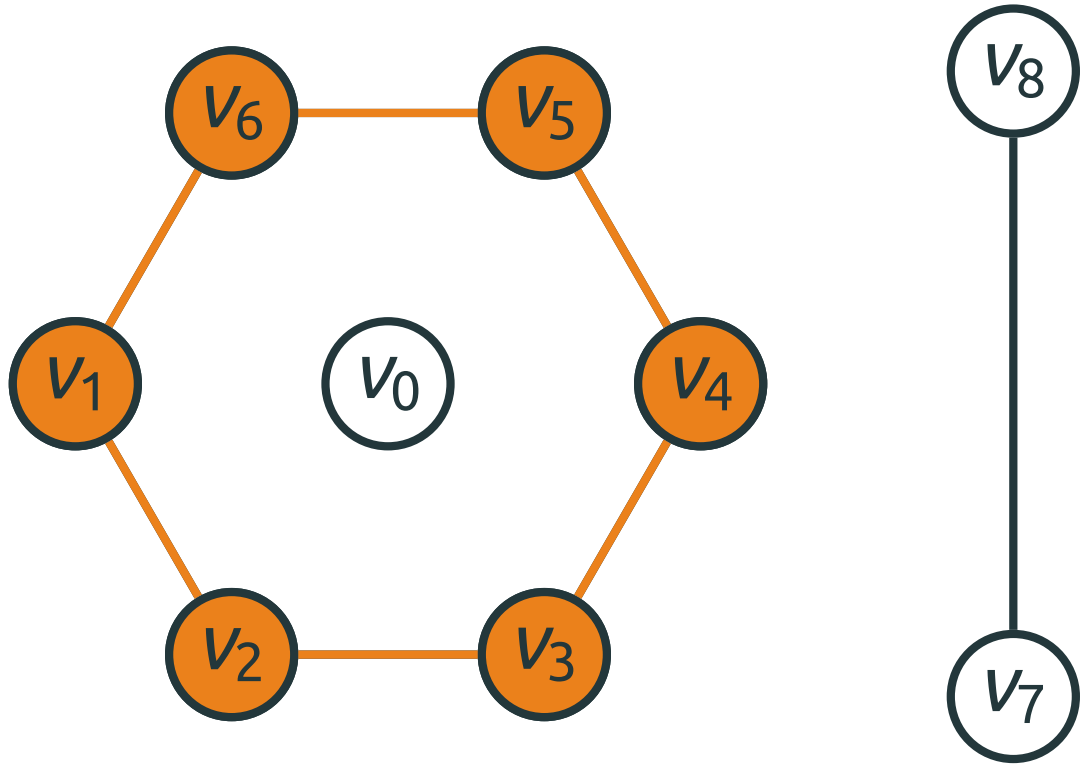
But not a connected component



Connected Components: Examples

v_1, v_2, v_3 form a connected subgraph

But not a connected component

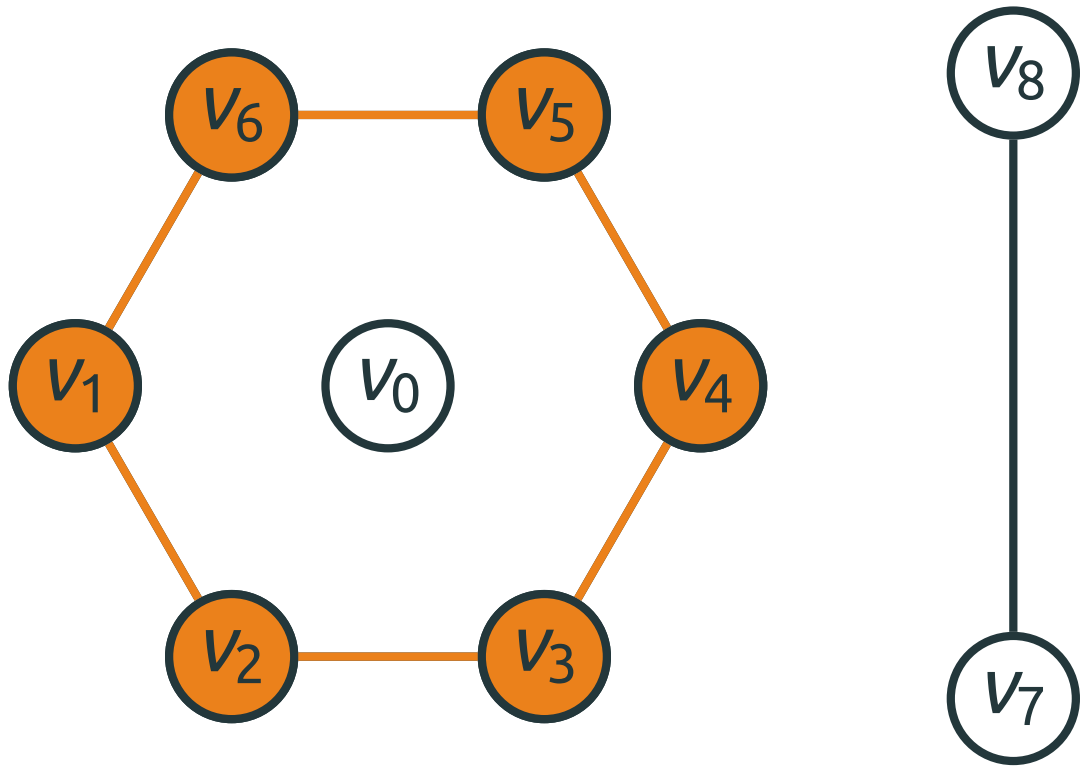


Connected Components: Examples

v_1, v_2, v_3 form a connected subgraph

But not a connected component

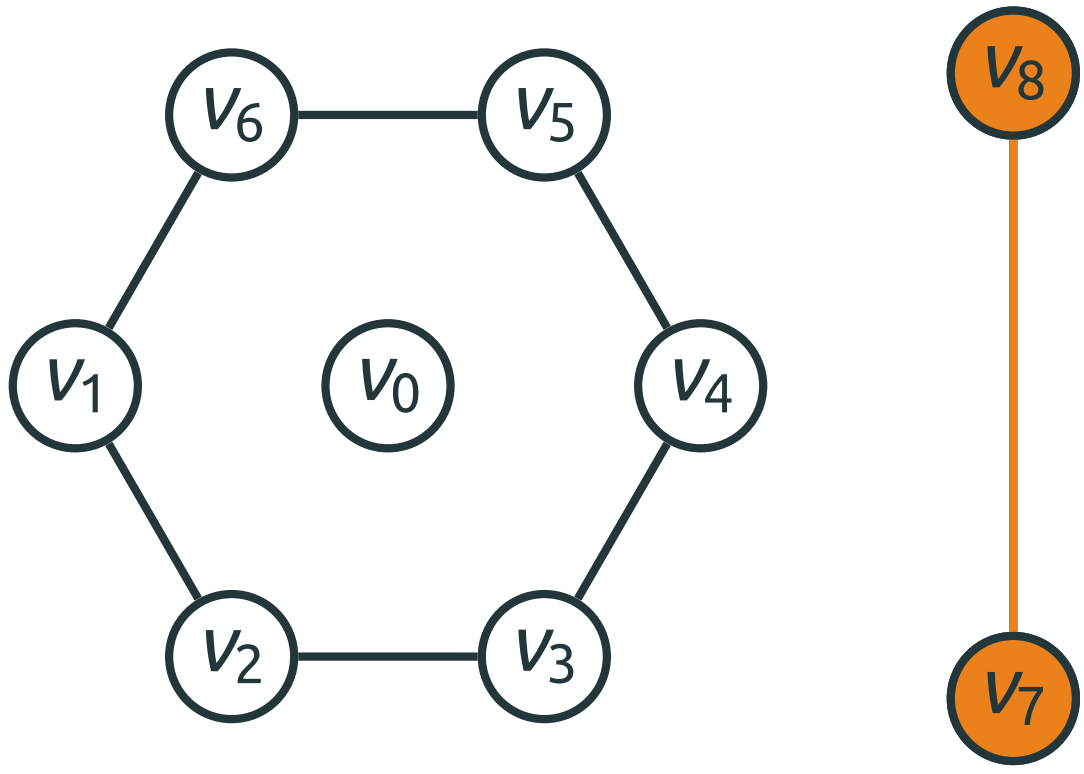
$v_1, v_2, v_3, v_4, v_5, v_6$ form a Connected Component



Connected Components: Examples

v_7, v_8 form a Connected Component

v_1, v_2, v_3, v_4, v_5, v_6 form a Connected Component

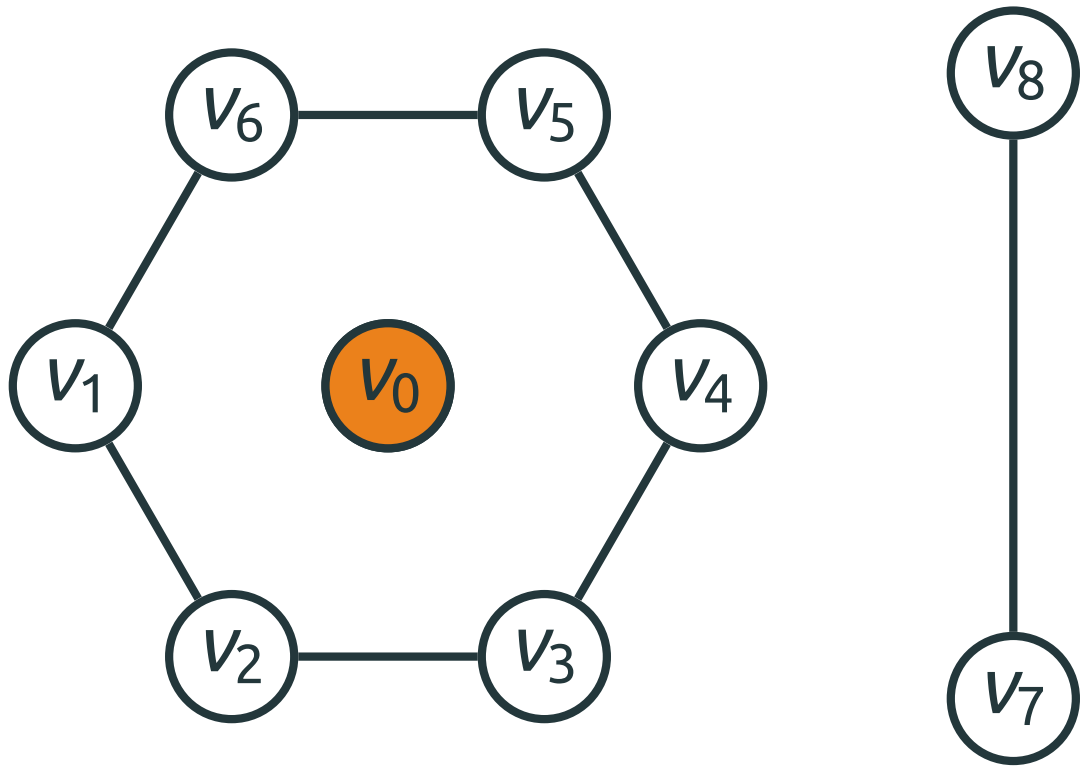


Connected Components: Examples

v_0 forms a Connected Component

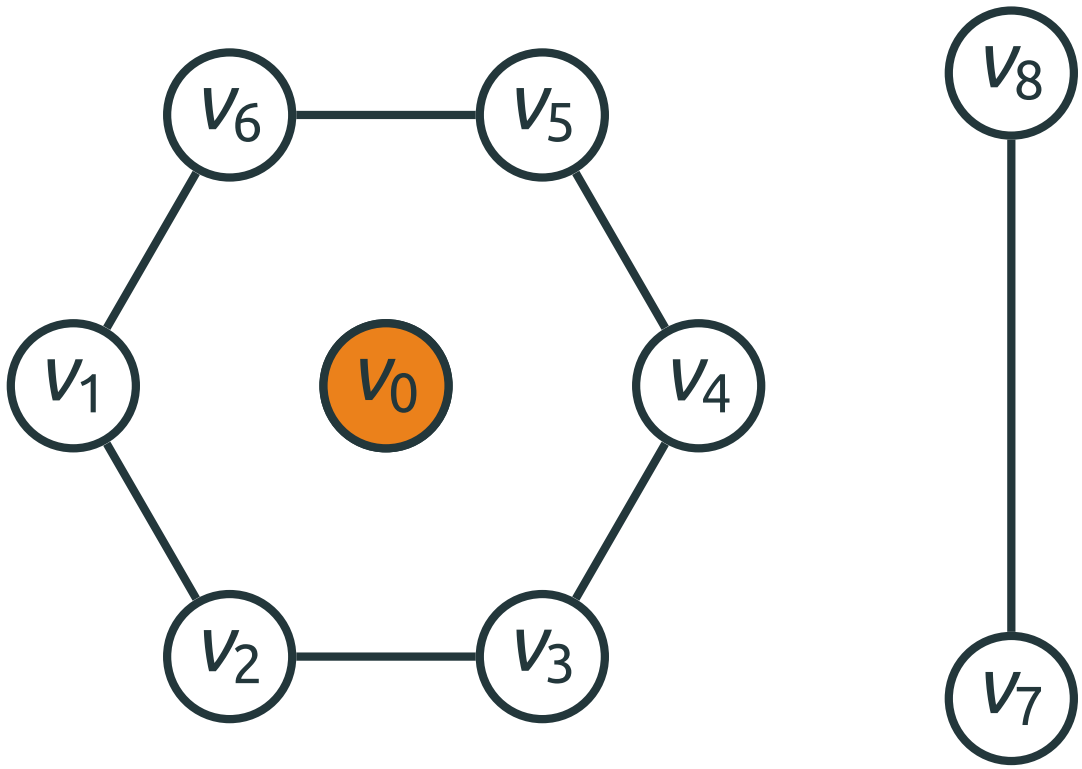
v_7, v_8 form a Connected Component

v_1, v_2, v_3, v_4, v_5, v_6 form a Connected Component



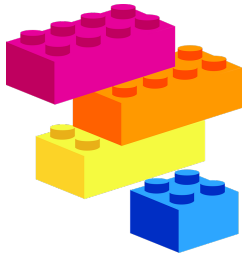
Connected Components: Examples

Each isolated vertex forms a Connected Component

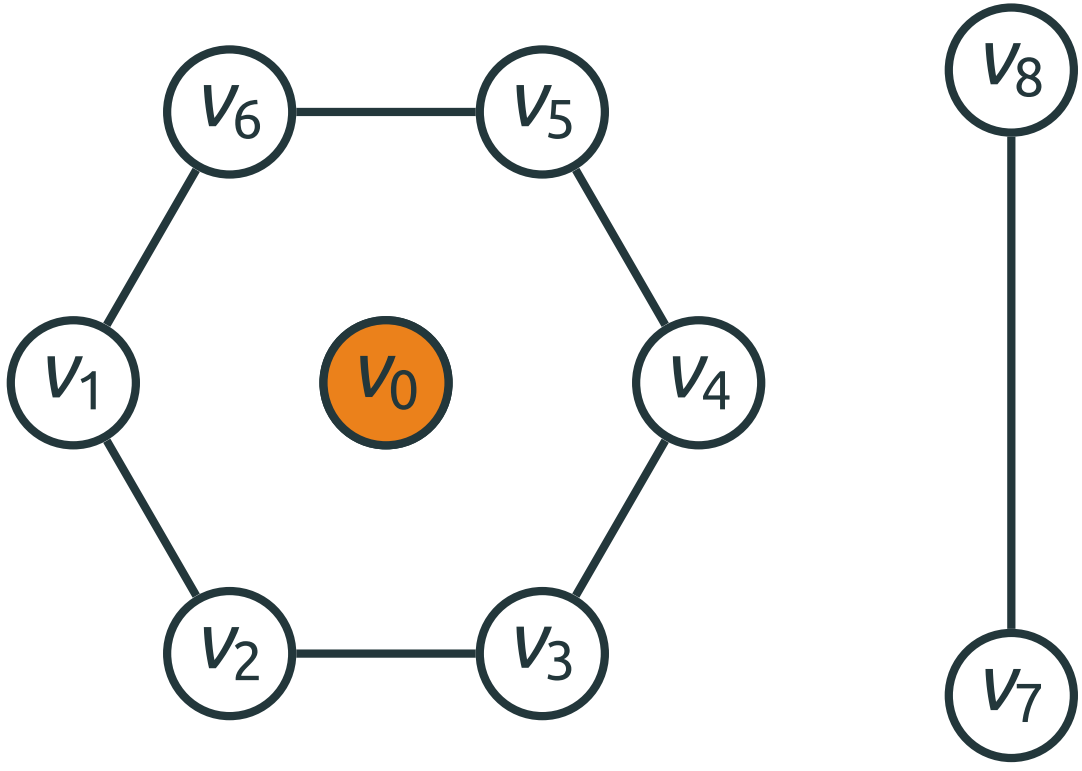


Connected Components: Examples



Each isolated vertex forms a Connected Component



Outline

The Degree of a Vertex

Paths

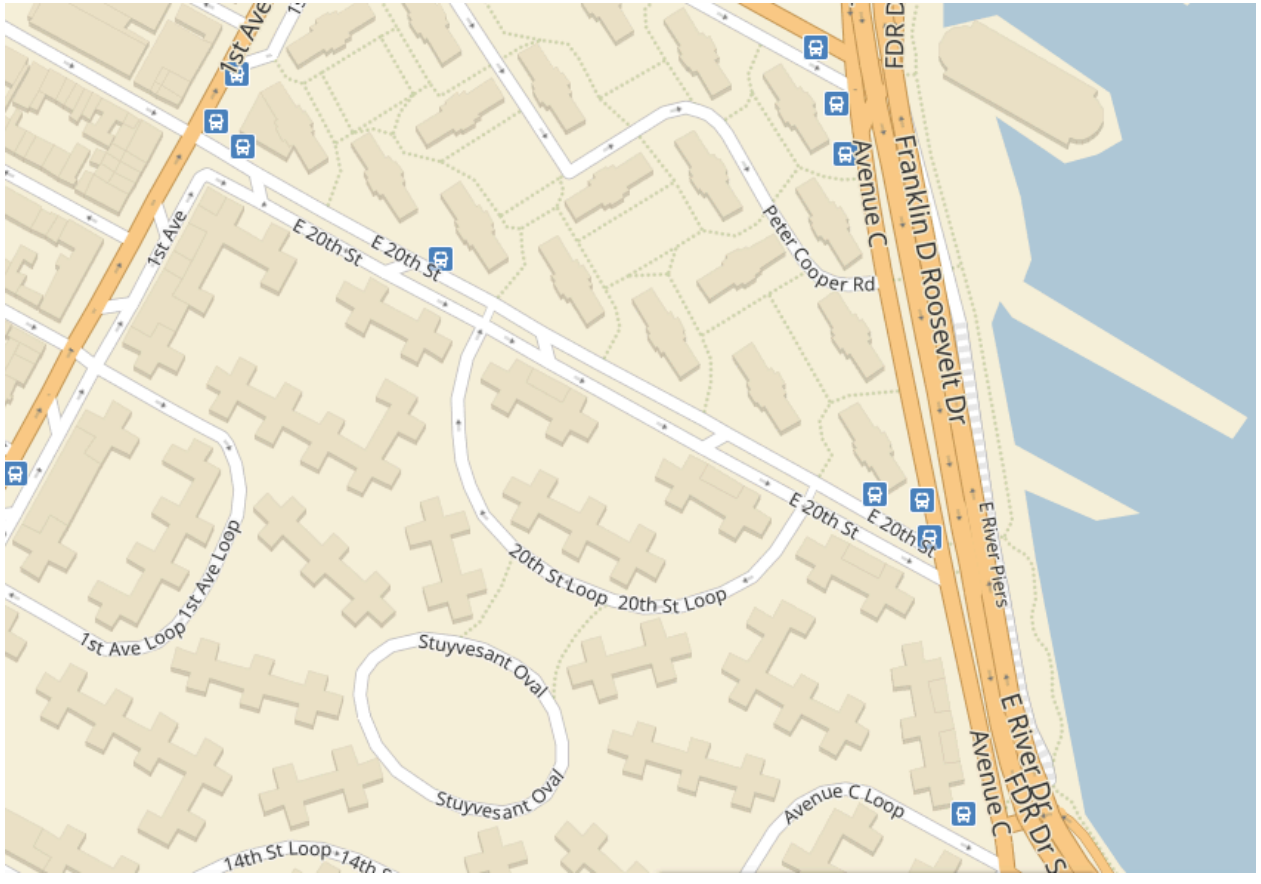
Connectivity

Directed Graphs

Weighted Graphs

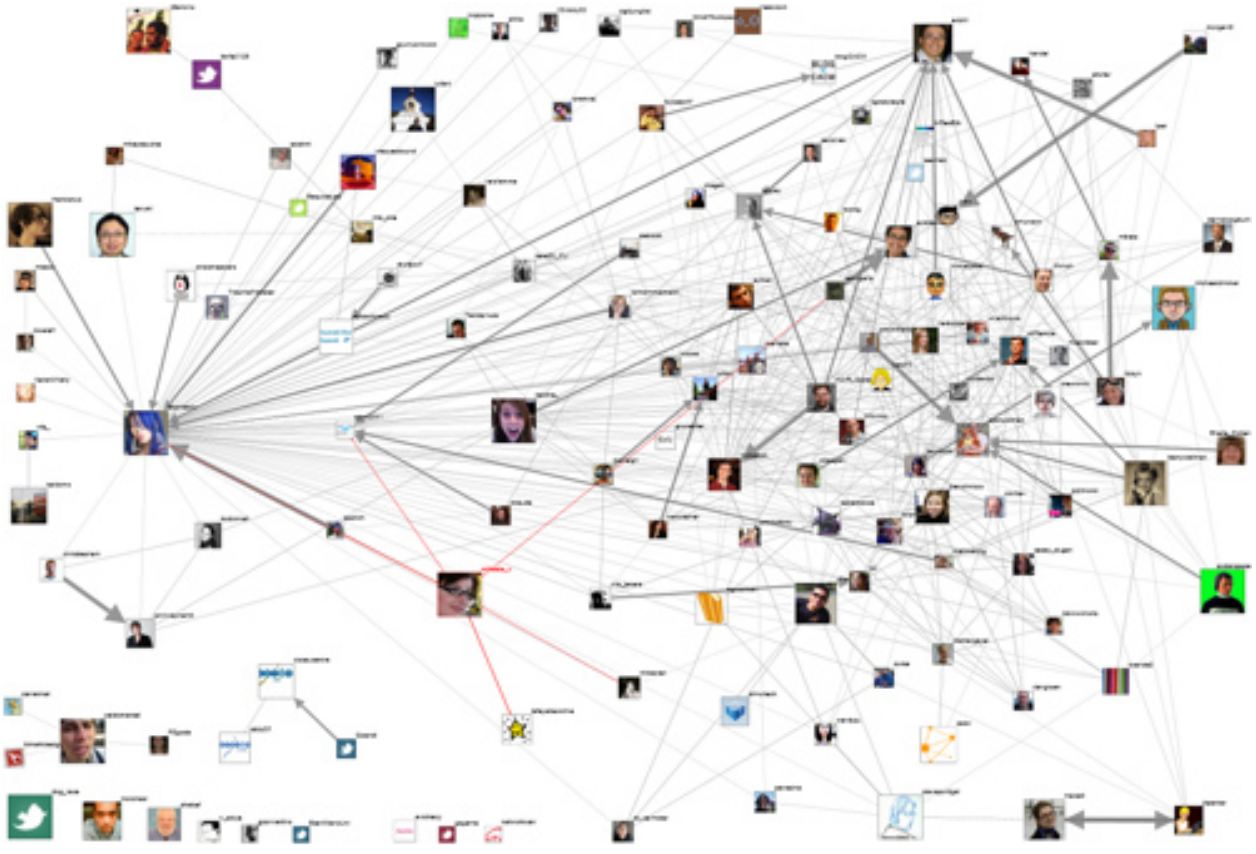
Directed Graphs

One-way Streets



Directed Graphs

Followers



Undirected Edge (Edge)

Edge $\{u, v\}$



Directed Edge (Arc)

Arc (u, v)



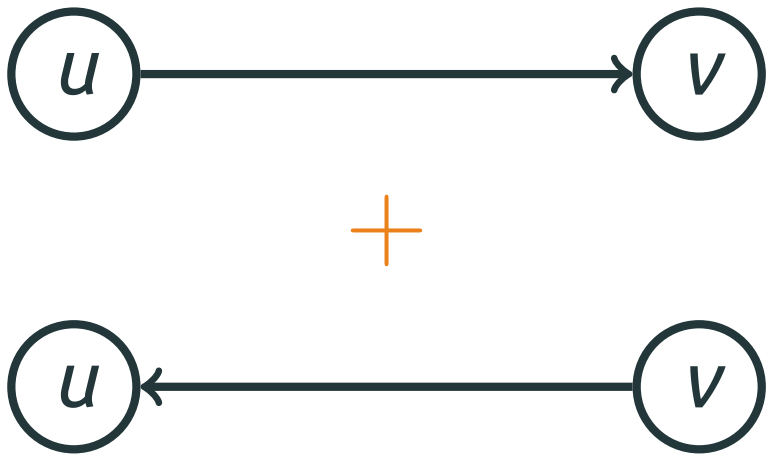
Directed Edge (Arc)

Arc (u, v)



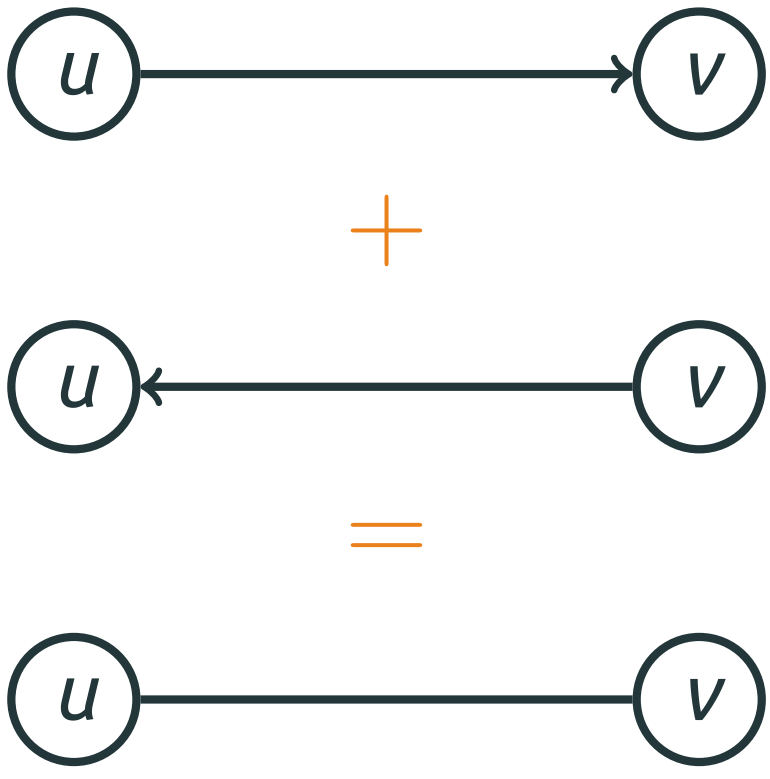
Directed Edge (Arc)

Arc (u, v)



Directed Edge (Arc)

Arc (u, v)



The Degree of a Vertex

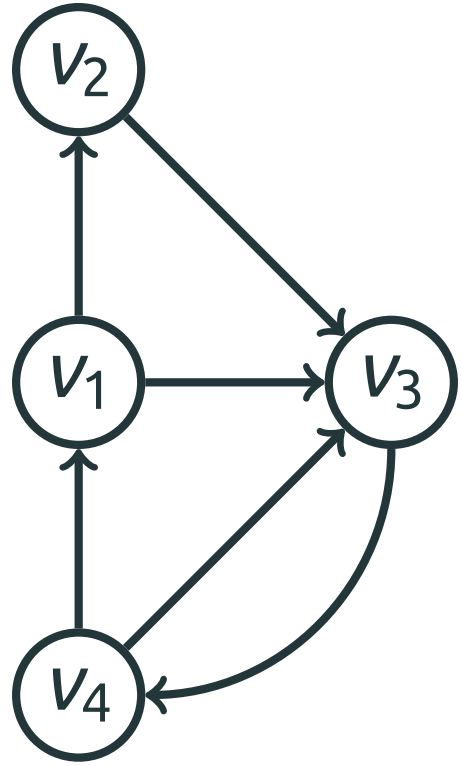
- The Indegree of a vertex v is the number of edges ending at v

The Degree of a Vertex

- The Indegree of a vertex v is the number of edges ending at v

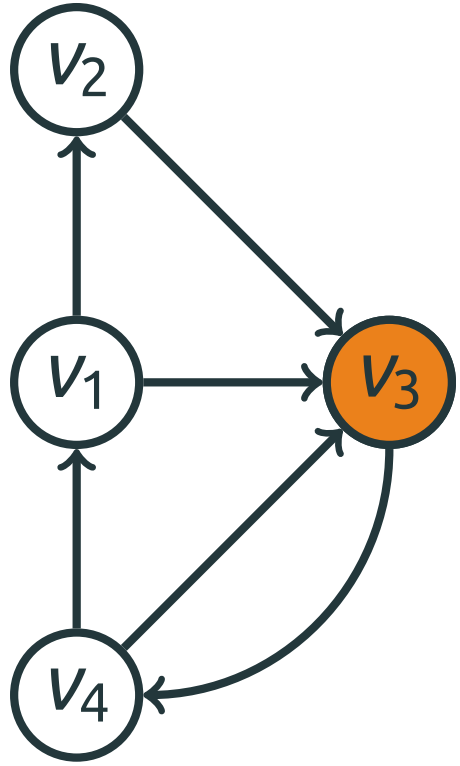
- The Outdegree of a vertex v is the number of edges leaving v

The Degree of a Vertex: Examples



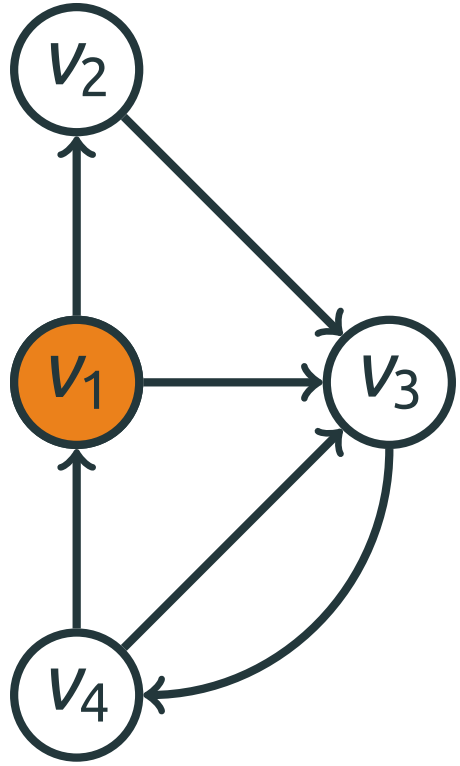
The Degree of a Vertex: Examples

The Indegree of v_3 is 3,
the Outdegree of v_3 is 1

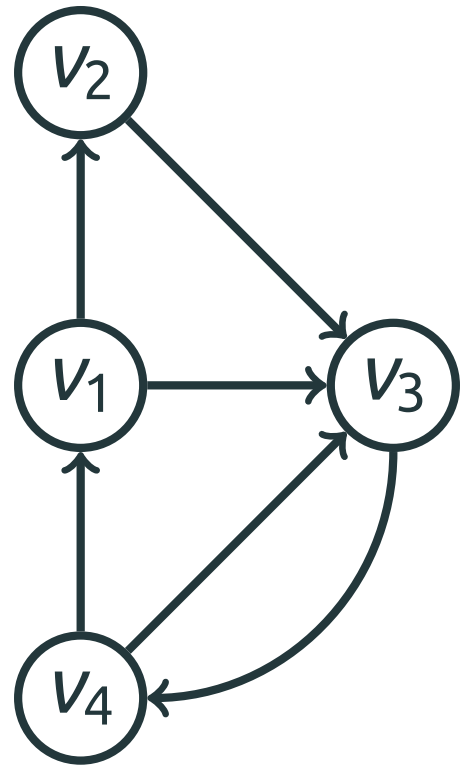


The Degree of a Vertex: Examples

The Indegree of v_1 is 1,
the Outdegree of v_1 is 2

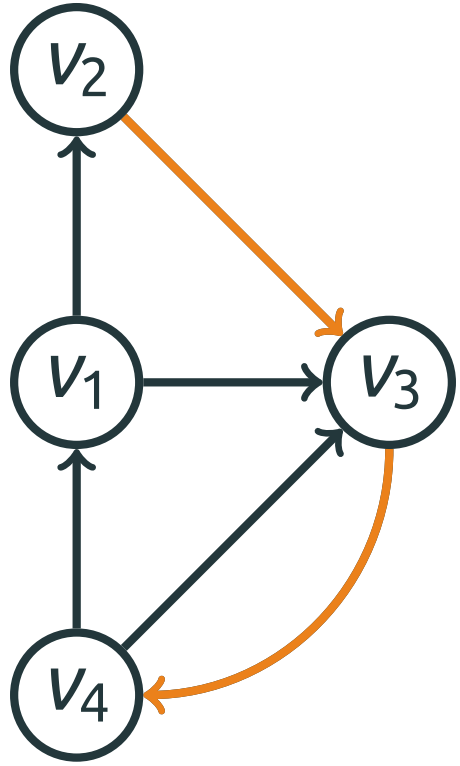


Directed Paths



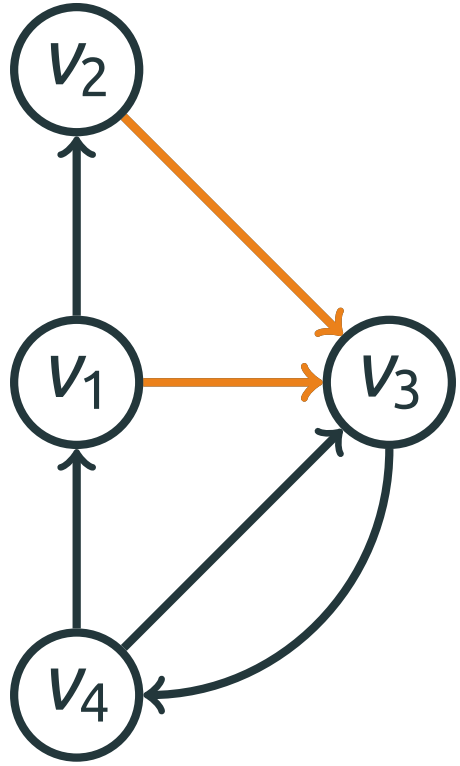
Directed Paths

(v_2, v_3, v_4) is a Path of length 2



Directed Paths

(v_1, v_3, v_2) is not a Path



Outline

The Degree of a Vertex

Paths

Connectivity

Directed Graphs

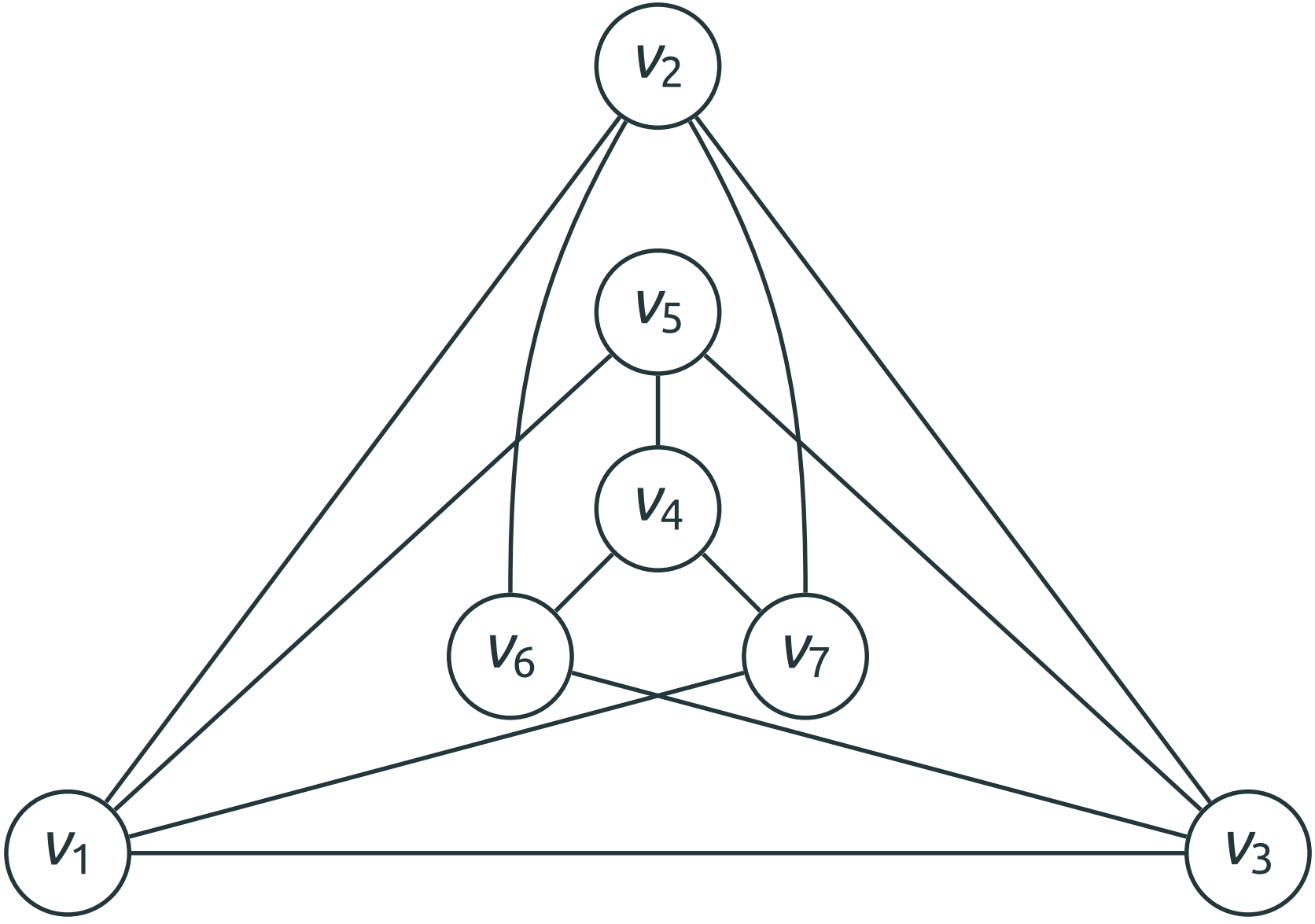
Weighted Graphs

Weighted Graphs

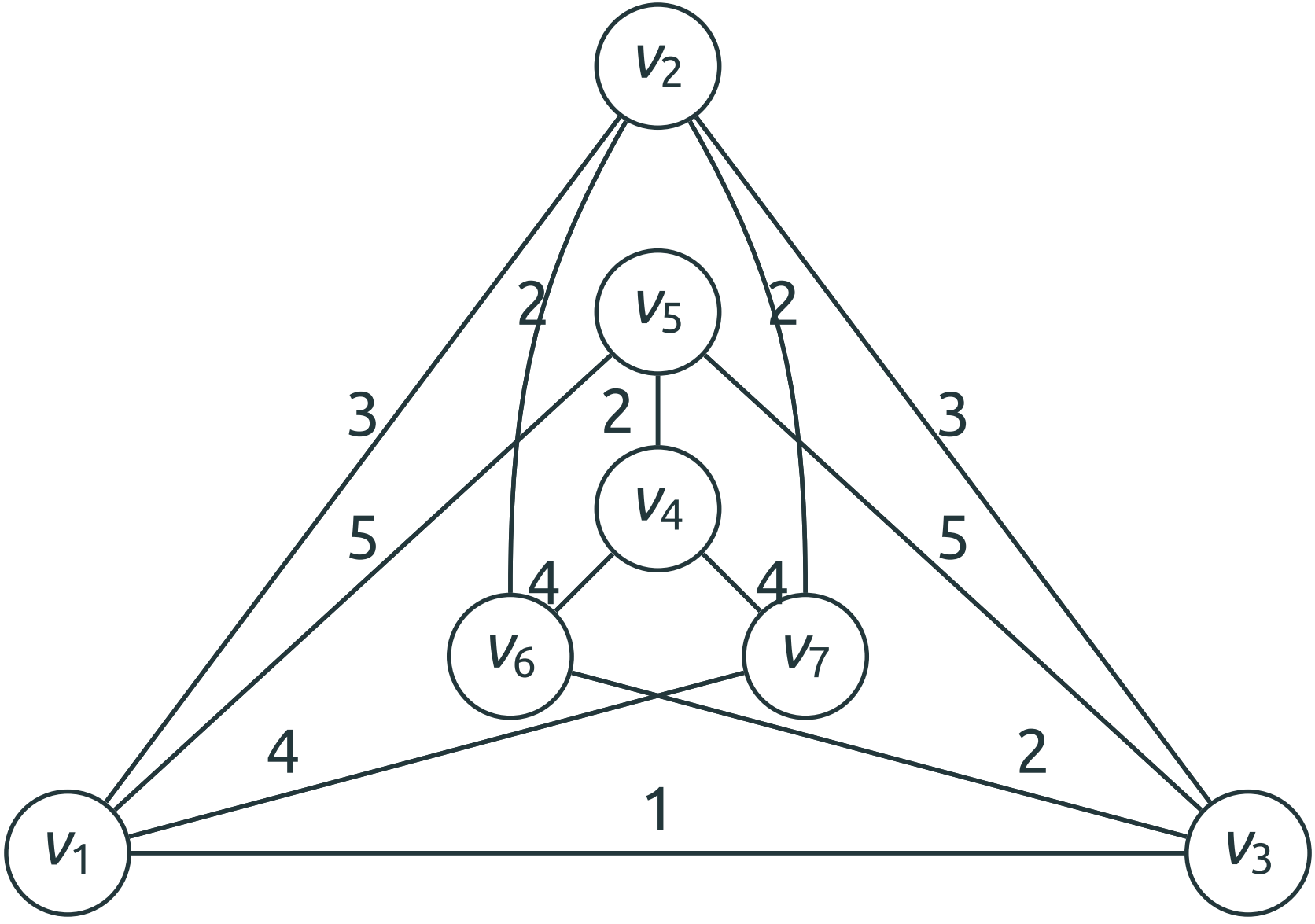
Distance, Driving Time, etc.



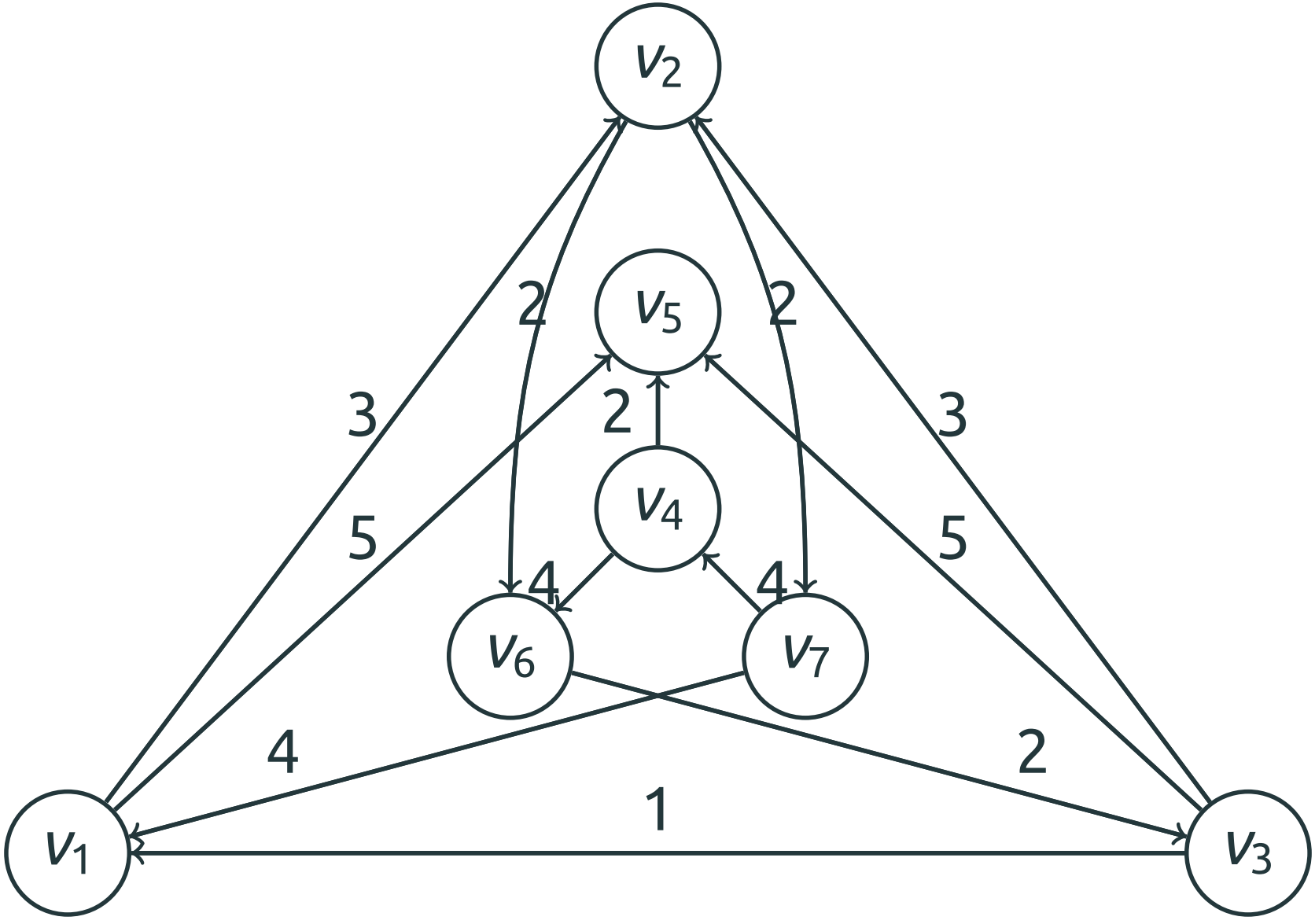
Weighted Graphs: Examples



Weighted Graphs: Examples



Weighted Graphs: Examples



Weighted Paths

- A Weighted Graph associates a weight with every edge

Weighted Paths

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- The Weight of a path is the sum of the weights of its edges

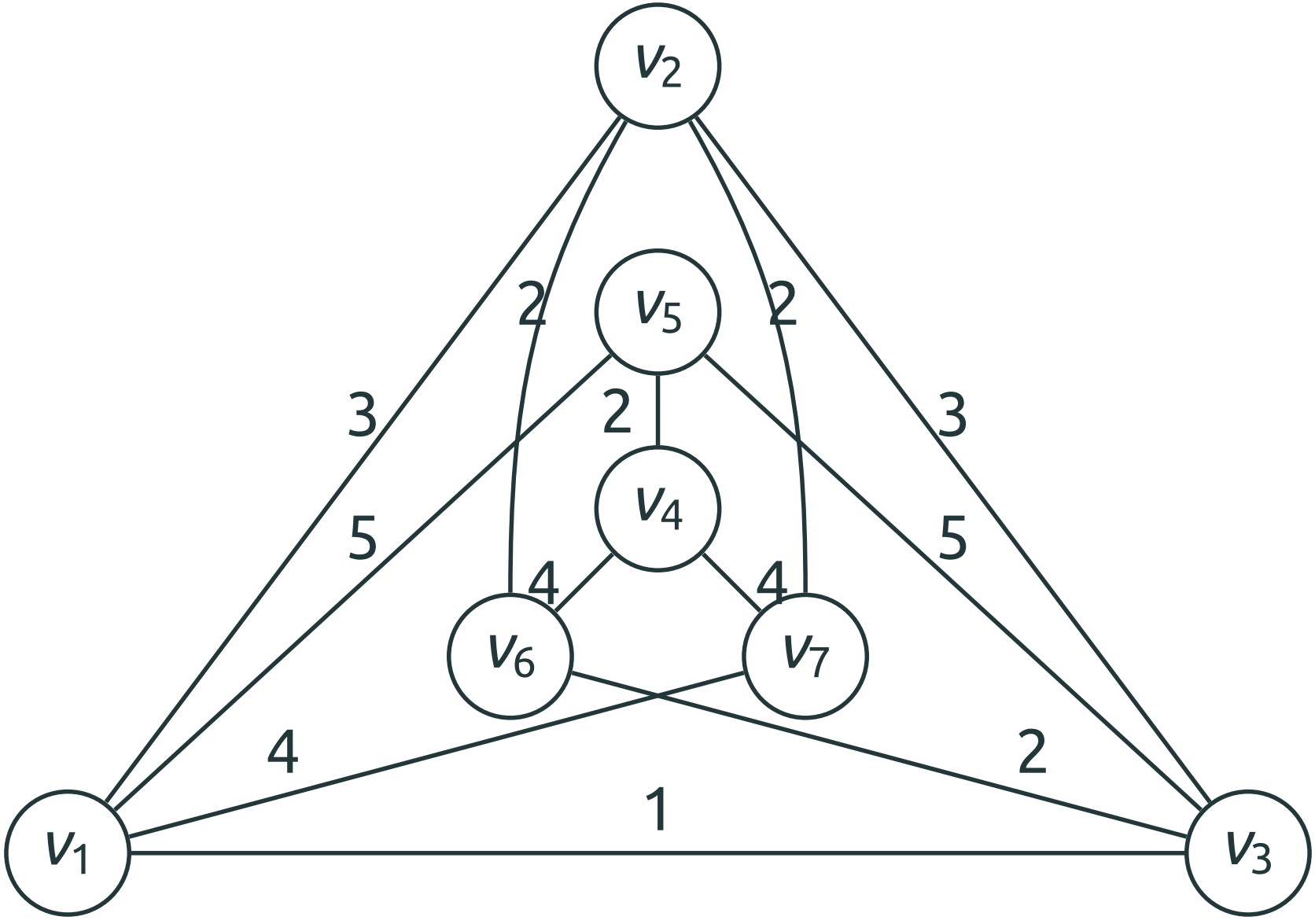
Weighted Paths

- A Weighted Graph associates a weight with every edge
- The Weight of a path is the sum of the weights of its edges
- A Shortest Path between two vertices is a path of the minimum weight

Weighted Paths

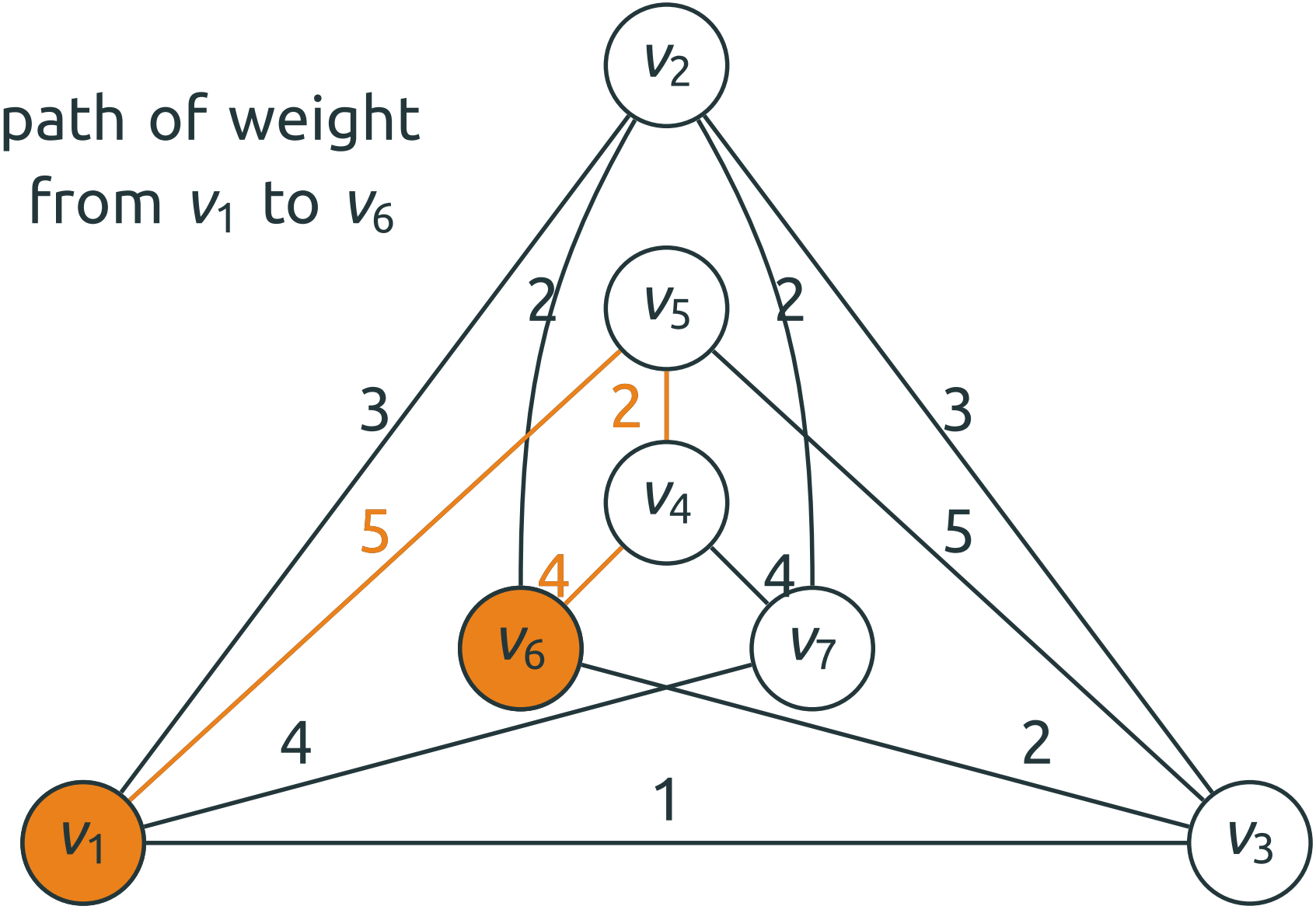
- A Weighted Graph associates a weight with every edge
- The Weight of a path is the sum of the weights of its edges
- A Shortest Path between two vertices is a path of the minimum weight
- The Distance between two vertices is the length of a shortest path between them

Weighted Paths: Examples



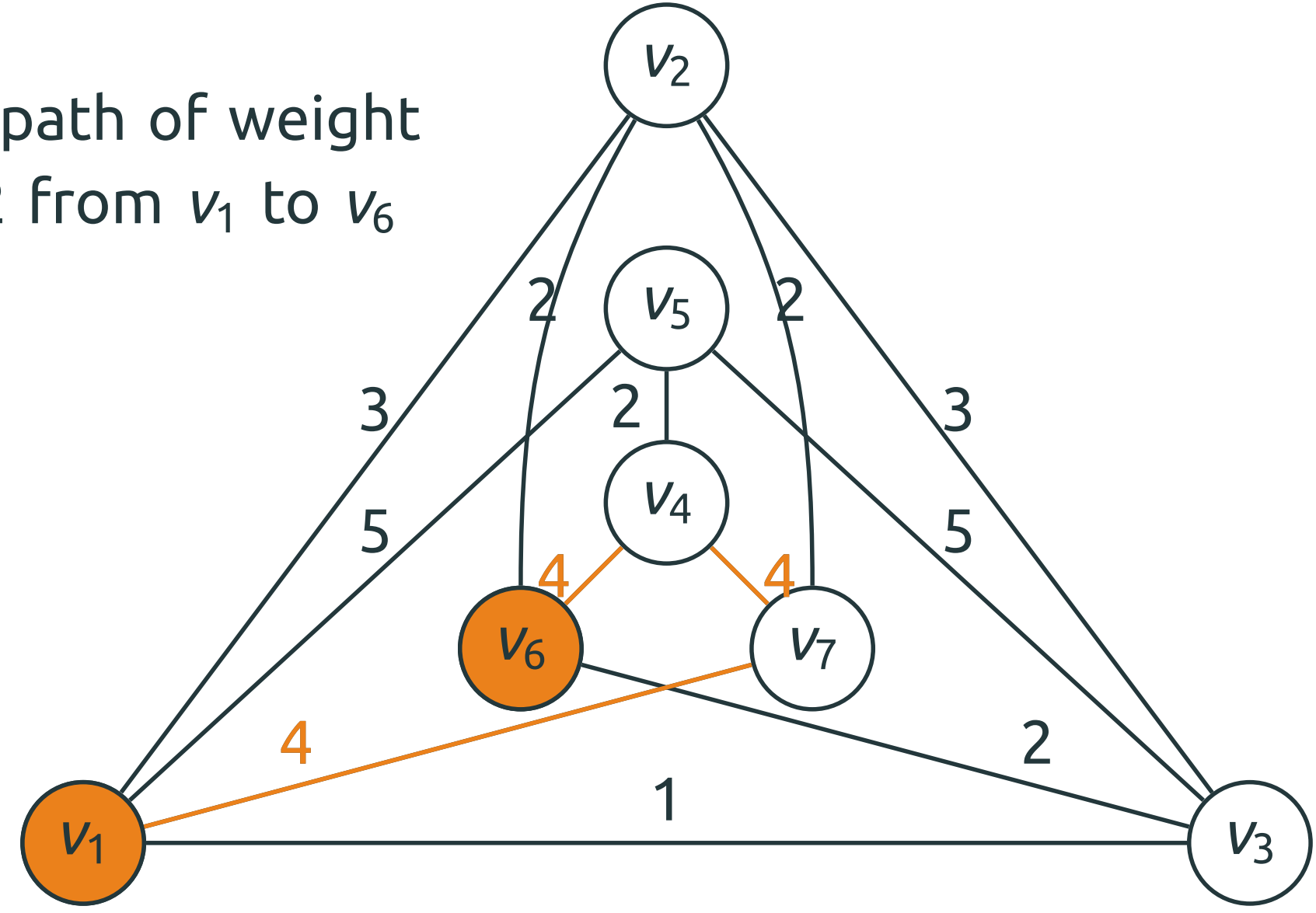
Weighted Paths: Examples

A path of weight 11 from v_1 to v_6



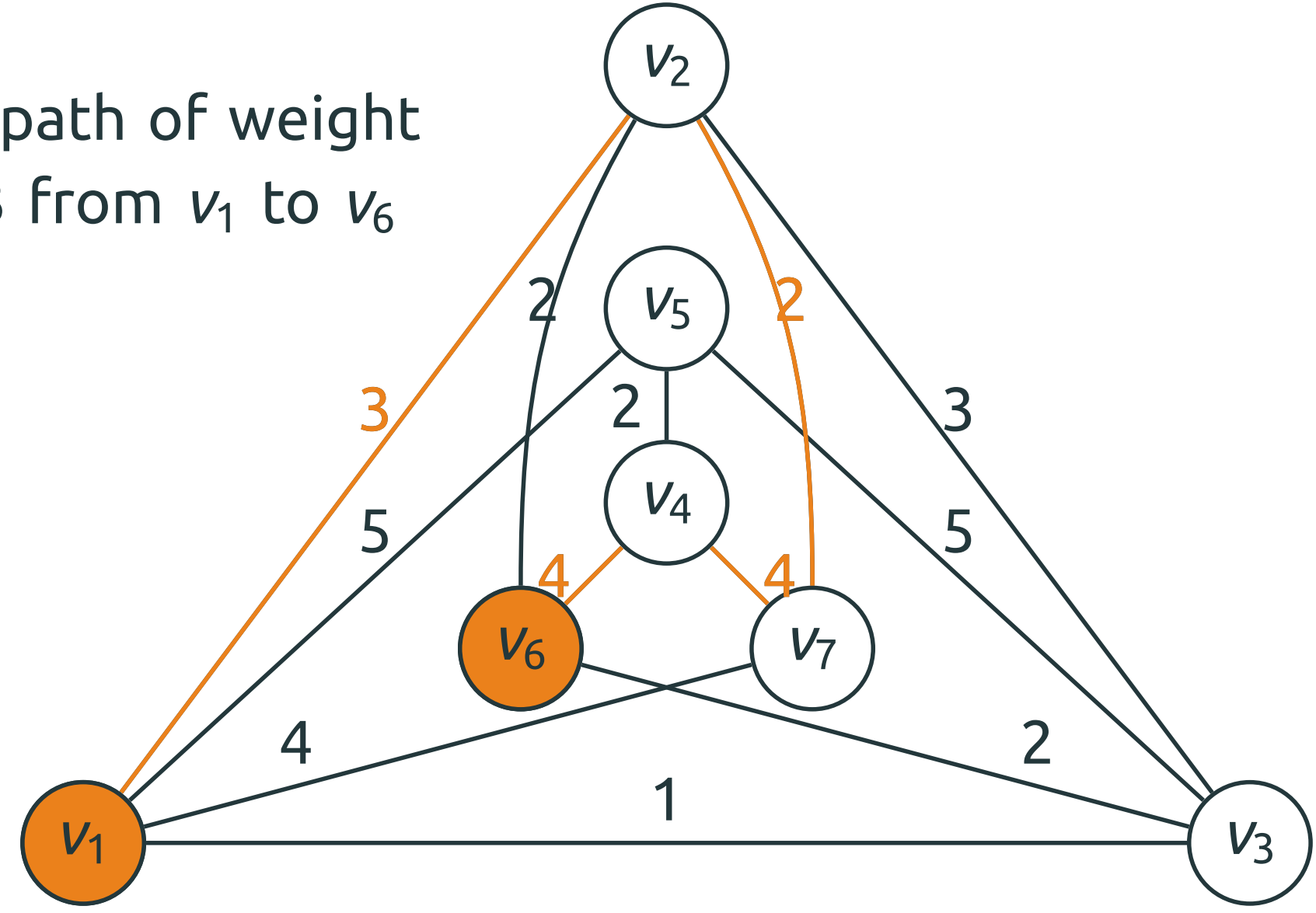
Weighted Paths: Examples

A path of weight 12 from v_1 to v_6



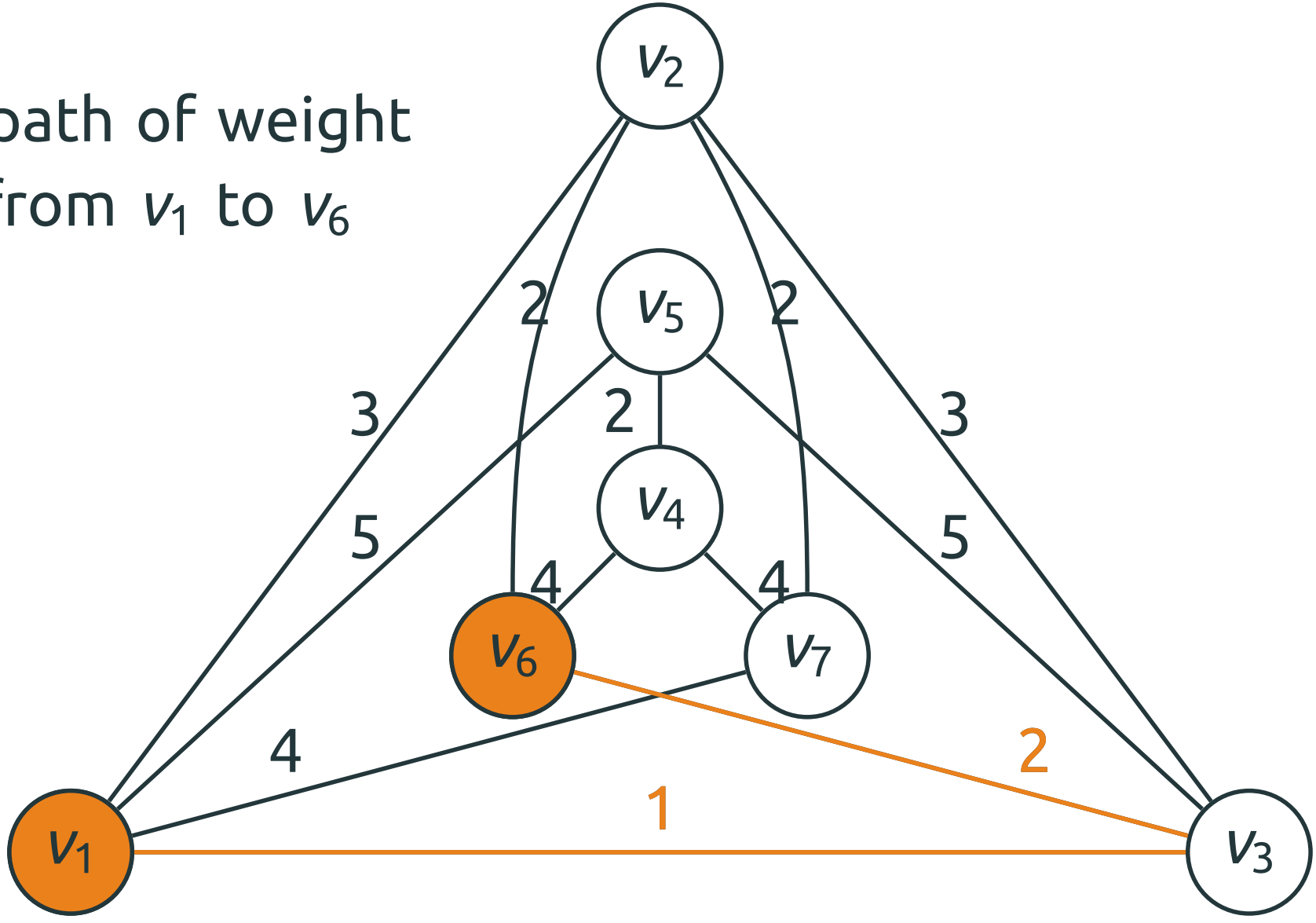
Weighted Paths: Examples

A path of weight 13 from v_1 to v_6



Weighted Paths: Examples

A path of weight 3 from v_1 to v_6



Weighted Paths: Examples

The distance between v_1 and v_6 is 3

